

THE ALGEBRAIC STRUCTURE OF MORPHOSYNTAX

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ABSTRACT. Within the context of the mathematical formulation of Merge and the Strong Minimalist Thesis, we present a mathematical model of the morphology-syntax interface. In this setting, morphology has compositional properties responsible for word formation, organized into a magma of morphological trees. However, unlike syntax, we do not have movement within morphology. A coproduct decomposition exists, but it requires extending the set of morphological trees beyond those which are generated solely by the magma, to a larger set of possible morphological inputs to syntactic trees. These participate in the formation of morphosyntactic trees as an *algebra over an operad*, and a *correspondence between algebras over an operad*. The process of structure formation for morphosyntactic trees can then be described in terms of this operadic correspondence that pairs syntactic and morphological data and the morphology coproduct. We reinterpret in this setting certain operations of Distributed Morphology as transformation that allow for flexibility in moving the boundary between syntax and morphology within the morphosyntactic objects.

1. INTRODUCTION

This work furthers the development of a mathematical formulation for generative linguistics, as initiated in [17] in the context of Chomsky’s Strong Minimalist Thesis, by extending the mathematical formulation of Merge and the core computational structure of syntax to another component of the language faculty—the morphology-syntax interface.

The morphology-syntax interface is linguistically conceptualized as how the internal structure of words relates to structures generated by syntax, and the extent to which the rules generating the former correlate to the rules constraining the latter [9]. We also interpret it as the ways that syntactic and morphological structures and processes combine in the assembly of morphosyntactic objects.

In order to explore the interface of syntax and morphology mathematically, the mathematical structure of morphology must first be established. The two prominent perspectives on morphology are Nanosyntax and Distributed Morphology (DM).

Nanosyntax [5, 25] stems from cartography and takes the perspective that the operations underlying syntax also underlie morphology. There is no distinct morphological system: morphological structures are the product of syntactic Merge. With respect to assembly of morphological components, there is a spellout loop comprised of syntactic assembly of a tree up to completion of a phase, then lexicalization of the section of the syntactic structure, before more syntactic structure-building and more spellout in the next phase, and so on. On the other hand, Distributed Morphology [10, 11] views morphology as the housing of morphological features in leaves of syntactic trees which can be manipulated by a variety of operations.

In this work, we take an intermediary perspective—morphological features end up as feature bundles in leaves of syntactic trees, but these bundles are actually hierarchical tree structures with morphological features at the leaves and feature bundles at internal nodes.

The model we present shares with Nanosyntax the idea that the underlying fundamental algebraic operations in morphology rely on the same computational structure that governs syntax, but with some specific adaptations that control and manipulate the feature hierarchies and the flexible boundary between morphology and syntax.

To avoid possible misunderstandings of our approach as presented in this paper, we begin by providing some general guidelines for how to interpret the construction presented in this paper. A presentation more focused on the linguistic interpretation and less detailed on the mathematical properties will be presented as a separate forthcoming paper, which will also help clarify some of these interpretations. Some important aspects to keep in mind while reading through the paper are listed here.

- *Our use of syntax, morphology, and morphosyntax terminology.* By syntax we mean the fundamental computational structure based on free symmetric Merge acting on workspaces, in the mathematical formulation developed [17]. By morphology we simply mean here the assembly of feature bundles, sometimes known in morphology as the theory of bundling. The underlying algebraic structure is fully inherited from syntax (it is the same free non-associative commutative magma operation underlying the formation of syntactic objects), and the only difference is the labeling procedure: while in syntax this is determined by a head function, and is a separate mathematical function distinct from the magma, a simpler union operation suffices for feature bundle formation, and the magma itself applies the label. By morphosyntax we then mean combination (via a geometric correspondence) of the computational structure of syntax and the feature bundles.
- *Computation is governed by syntax.* The mathematical model of Merge developed in [17] shows that a very constrained algebraic structure describes the structure formation via free symmetric Merge. The two key components of this structure are the magma operation and a coproduct operation that allows for the extraction of accessible terms as computational material. Other parts of the faculty of language, such as the syntax-semantics interface, also discussed in [17], are syntax-driven, and rely on these same computational mechanisms. The situation with morphology is similar: again the magma operation and the coproduct operation that exist in syntax are the key computational structures (in this sense, it is *syntax all the way down*), and the interface relies then simply on a correspondence matching lexical items and syntactic features to feature bundles. However, this correspondence does not mean that the syntactic tree assembly is not occurring with morphological objects at the leaves—rather, one can conceptualize the lexical items and syntactic features as being “placeholders” for the morphological objects. The existence of the correspondence does *not* mean that with respect to the timing of the derivation steps, the derivation is happening with objects that are not morphological in nature. The next point discusses this notion of timing of derivation versus computational components further.
- *Decoupling of independent computational substructures versus derivational steps.* A key idea in Chomsky’s recent (starting around 2013) formulation of Minimalism is the decoupling of all parts of the computational structure that can be decoupled. This idea led directly to identifying a free symmetric Merge as the core computational mechanism, interfaced with a filtering system (for well formed structure, in terms of phases, theta role assignments), necessary for parsing at the syntax-semantics interface, and further language specific filters at Externalization (parametric variation). In this description of syntax, one can take two different, but provably equivalent,

viewpoints. The key to understanding this fact is that the separation of independent computational components is not a separation in terms of time ordering of operations in a derivation, but rather a separation of their conceptual algebraic structures. This decoupling of independent computational modules has the advantage of rendering the algebraic structure transparent and making it easy to prove general results (which help to reduce the number of assumptions of the model and select between alternatives). On the other hand, efficiency of algorithmic realizations prefers a viewpoint in which the filtering is done step by step along structure formation. While this may appear at odds with the decoupling of structure formation and filtering, in fact these two perspective are fully equivalent (this equivalence is proven in [20] and [18]). The situation we present here in describing morphology and what we refer to as the morphology-syntax interface (and morphosyntax) is an analogous situation, where for the sake of clarity in the identification of the key algebraic structures, we decouple all the substructures that can be decoupled and describe them individually along with their interface mechanism. This again should not be read as a separation, at the level of time-ordered sequences of derivations, between syntax and morphology: there is no such thing, as the core computational structure is always driven by syntax.

- *Operations of Distributed Morphology.* Finally, the algebraic structures we identify provide a mathematical formulation for the fusion, fission, obliteration and impoverishment operations of Distributed Morphology. In view of the nature of the core computational mechanism (the magma and coproduct operations), this shows that it is fusion and fission that form the key basic operations. Fusion and fission switch between the two modes of labeling of the (syntactic) magma, unlabeled (syntax) and labeled (morphological feature bundle formation), and therefore flexibly moving the syntax-morphology boundary, that is, the boundary between the unlabeled syntactic nodes and the labeled, by either individual features or feature bundles, morphological nodes. The obliteration and impoverishment operations, which are often presented as the fundamental ones in Distributed Morphology, are more naturally seen (and mathematically provable) as derived from the fusion/fission ones and the coproduct. Not only is this a striking result from the morphological perspective, but mathematically it again demonstrates a reduction of the number of required operations and a simplification of the overall computational structure of the system.

As with the syntactic objects, the morphological tree structures are generated by a magma operation. The two algebraic structures of syntax and morphology begin to differ at the level of workspaces and their algebraic structure. In syntax, in the model of [17], workspaces are forests of syntactic objects, and the vector space they span is endowed with a Hopf algebra structure, with the action of Merge realized as a Hopf algebra Markov chain. In the case of morphology, one can also form workspaces with a coproduct operation that allows for extraction and elimination of parts of feature bundles, but this operation requires an extension of the underlying space beyond the objects formed by the magma, which is obtained via a comodule structure. The morphological objects obtained in this way then provides the inputs at the leaves of syntactic trees. This insertion of morphological structures at the leaves of syntactic trees is realized algebraically as an algebra over an operad, as suggested in [17] and in [19], but with a subtlety: we need to introduce a more refined notion of *correspondence between algebras over an operad*. Moreover, a main difference with respect to syntax is that, instead of having a Merge operation acting as a Hopf algebra Markov

chain on the morphological workspaces, as with the syntactic space, one has a more complex structure formation operation that acts only at the level of morphosyntactic trees. Since in these trees the boundary between syntax and morphology is somewhat flexible (in a way that is made precise by two operations, fusion and fission, that push the boundary either upward or downward, respectively), this action can sometimes be interpreted in syntactic or in morphological terms.

Thus, the morphology-syntax interface in this work is described by a number of mathematical systems operating partly sequentially and partly in parallel and interfacing with each other, whose mathematical structures are not identical but must be compatible with each other to the extent that they must compose to form, and operate on, the final morphosyntactic structures. It should be pointed out that this entire model is pre-Externalization. Externalization then acts on the structures obtained in this way as a selection process governed by *morphosyntactic parameters* that select viable structures in a language-dependent way and incorporates language-specific lexicon. Then, in parallel to the process in syntax, any ill-formed structures will be filtered out via morphosyntactic parameters. One of these will be compatibility with Feature Geometry [13]. We will not discuss the Externalization part and the parametric variation in this paper, where we only focus on the underlying computational structure that acts in a way that is independent of realization in a specific language.

The following is a quick summary of the various components of the Morphology-Syntax interface, and their consecutive order.

- (1) Syntactic and morphological workspaces exist simultaneously and independently:
 - (a) *Syntax*: The workspaces are forests whose components are syntactic objects (the elements of the free non-associative commutative magma generated by the set of lexical items and syntactic features). The linear span of workspaces has a coproduct that extracts accessible terms for structure formation and movement realized by Merge. The Merge action on workspaces has the form of a Hopf algebra Markov chain with respect to the coproduct that extracts accessible terms and the product that forms workspaces. This is the mathematical model of Merge developed in [17]: it will be briefly summarized in the following section.
 - (b) *Morphology*: Morphological trees are also built by a free non-associative commutative magma, as binary tree structures with morphological features (some of which are valued, some of which are not) at the leaves. Workspaces now include components that are not just the elements of this magma (unlike syntax) but include additional objects generated by extraction and elimination of features via the coproduct. The resulting linear space of these extended morphological workspaces then has a resulting Hopf structure with product and coproduct.
- (2) *Morphosyntactic trees* are obtained as insertion of morphological trees at the leaves of syntactic trees in a way that satisfies a compatibility rule between the syntactic features labeling the leaves of the syntactic trees and the morphological feature bundles labeling the roots of the morphological trees. Syntactic objects form an algebra over an operad (as shown in [17] and in [20]), and the correct mathematical structure that describes this insertion of morphological trees into syntactic trees is identified as a correspondence of algebras over an operad.
- (3) *Morphosyntactic workspaces* are forests whose components are morphosyntactic trees, where the morphological structures inserted at the leaves of the syntactic trees also

include components of morphological workspaces that are not in the magma of morphological trees. The syntactic trees can then be understood as defining a family of operations that map morphological to morphosyntactic workspaces. These operations are formally similar to the Hopf algebra Markov chain of Merge acting on syntactic workspaces, but rely on already built syntactic objects.

- (4) *Distributed Morphology*: post-syntactic morphological operations of fission, fusion, impoverishment, and obliteration now can be seen as transformations of morphosyntax trees. They are “post-syntactic” in the sense that they act after syntactic trees are interfaced with morphological trees in the formation of morphosyntactic objects, but are still acting prior to Externalization.

While we recognize that DM is not universally agreed upon as the computational mechanism underlying morphology, the purpose of including this here is to demonstrate that the system is capable of handling post-syntactic morphological operations. Moreover, DM is in particular interesting as it has the overall effect of moving the boundary between syntax and morphology in the morphosyntactic trees.

In addition to these aspects of the model that we will be developing in detail in the rest of the paper, there are other directions, which we will not be including here but that we expect will also have a mathematical formulation compatible with this model. These include:

- *Agree*: valuation of unvalued features in the morphological trees via another colored operad similar to the treatment of theta theory and phases in [20] and [18].
- *Externalization: Planarization and Filtering*: in the model we develop here all trees are non-planar. The choice of a planar structure is part of Externalization (and is language-dependent, for example in the structure of prefixes and suffixes in word formation) as in the case of the planarization of syntactic trees. In the same Externalization phase some of the freely formed structures are filtered out (again in a language-dependent way) according to morphosyntactic parameters.
- *Externalization: Vocabulary insertion*: after all morphological features have attained their final values, and the morphosyntax trees are in their final configurations, as part of the Externalization process language-specific morphemes are inserted at the leaves of the morphosyntax trees via a colored operad, and word formation takes place according to the features and the structure of the morphological trees.

In the next sections, we will elaborate on each of the individual components of the pre-Externalization model listed above.

1.1. Recalling the mathematical structure of syntax. In [17] a mathematical framework for syntax was developed, where the main algebraic structure is a Hopf algebra coproduct, responsible for the extraction of accessible terms from syntactic objects that is needed for the core generative procedure underlying the compositional structure of syntactic trees, namely the Merge operation. A brief summary of how Merge operates can be articulated in the following way:

- Syntactic objects are built by successive iterations of a non-associative commutative binary operation $\mathfrak{M}$, starting from a finite set $\mathcal{SO}_0$ of lexical items and syntactic features. The resulting set $\mathcal{SO}$ of lexical items is the free non-associative commutative magma generated by $\mathcal{SO}_0$,

$$\mathcal{SO} = \text{Magma}_{na,c}(\mathcal{SO}_0, \mathfrak{M})$$

and as such it can be identified with the set $\mathfrak{T}_{\mathcal{SO}_0}$ of non-planar full binary rooted trees with leaves decorated by elements of $\mathcal{SO}_0$.

- Merge is a dynamical system acting on workspaces: it takes a workspace as input and it outputs a sum of possible resulting workspaces (all the structures obtainable from the input in a single Merge move). Structure formation and movement are achieved by iteration of this Merge action. At each step, the current workspace that Merge is acting on is likened to a scratchpad, where steps of derivations take place, transforming the workspace, starting with an unstructured collection of lexical items, until a final completed sentence structure is obtained. At each step a workspace is a disjoint union (a forest) $F = \sqcup_a T_a$ whose components $T_a \in \mathcal{SO}$ are syntactic objects.
- The vector space $\mathcal{V}(\mathfrak{T}_{\mathcal{SO}_0})$ spanned by this set of forests (workspaces) $\mathfrak{T}_{\mathcal{SO}_0}$ has a product operation $\sqcup$ that combines two workspaces into a single one (and in particular places syntactic objects into a workspace) and a coproduct operation that extracts all the available material for computation in a workspace, namely all the accessible terms $T_v \subset T$ of components of the workspace, with T_v the full subtree of T below one of the vertices. The coproduct takes the form

$$(1.1) \quad \Delta(T) = \sum_{\underline{v}} F_{\underline{v}} \otimes T/F_{\underline{v}},$$

and with $\Delta(F) = \sqcup_a \Delta(T_a)$ for $F = \sqcup_a T_a$, where $\underline{v} = \{v_1, \dots, v_r\}$ are vertices with non-overlapping accessible terms T_{v_i} and $F_{\underline{v}} = T_{v_1} \sqcup \dots \sqcup T_{v_r}$ is the extracted material, with $T/F_{\underline{v}}$ the corresponding cancellation of the deeper copies.

- the Merge action on workspaces is then formulated as a composition of operations

$$(1.2) \quad \mathfrak{M}_{S,S'} = \sqcup \circ (\mathfrak{B} \otimes \text{id}) \circ \delta_{S,S'} \circ \Delta,$$

where (reading from right to left with $\circ$ indicating composition of functions)

- (1) first the coproduct Δ extracts all accessible terms making them available to be used for structure formation.
- (2) then $\delta_{S,S'}$ searches over all the extracted terms in the coproduct for a specific pair of syntactic objects S and S' to act on, and eliminates all terms that are not of the form $S \sqcup S' \otimes F'$ for some forest F' .
- (3) the remaining terms are acted upon by the operator $(\mathfrak{B} \otimes \text{id})$ resulting in terms of the form $\mathfrak{B}(S \sqcup S') \otimes F'$, where the operator $\mathfrak{B}$ grafts a forest to a common root

$$\mathfrak{B} : S \sqcup S' \mapsto \widehat{S \sqcup S'}$$

- (4) finally $\sqcup$ reassembles the new workspace $\mathfrak{B}(S \sqcup S') \sqcup F'$.

- this action of the Merge operations $\mathfrak{M}_{S,S'}$ of (1.2) gives rise to three possible cases: External Merge, Internal Merge, and Sideward Merge.

- (1) *External Merge*: $\mathfrak{M}_{S,S'}$ where S and S' are syntactic objects that are connected components of the workspace, $F = S \sqcup S' \sqcup \hat{F}$, resulting in a new workspace $\mathfrak{M}_{S,S'}(F) = \mathfrak{M}(S, S') \sqcup \hat{F}$ (where $\mathfrak{M}(S, S') = \mathfrak{B}(S \sqcup S')$).
- (2) *Internal Merge*: $\mathfrak{M}_{S,T/S} \circ \mathfrak{M}_{S,1}$ where 1 is the unit of the magma (the formal empty tree) where S is an accessible term $S = T_v$ of a component T of the workspace $F = T \sqcup \hat{F}$. Here $\mathfrak{M}_{S,1}$ has the effect of extracting T_v and placing it in the workspace and $\mathfrak{M}_{S,T/S}$ then merges it with the remaining structure T/S resulting in a new workspace $\mathfrak{M}(T_v, T/T_v) \sqcup \hat{F}$.

- (3) *Sideward Merge* has three cases: (a) $S = T_v \subset T$ an accessible term and $S' = T'$ a component of the workspace $F = T \sqcup T' \sqcup \hat{F}$, resulting in $\mathfrak{M}(T_v, T') \sqcup T/T_v \sqcup \hat{F}$; (b) $S = T_v \subset T$ and $S' = T_w \subset T$ two disjoint accessible terms of the same component T , resulting in $\mathfrak{M}(T_v, T_w) \sqcup T/(T_v \sqcup T_w) \sqcup \hat{F}$; (c) $S = T_v \subset T$ and $S' = T'_w \subset T'$ accessible terms of two different components, resulting in $\mathfrak{M}(T_v, T'_w) \sqcup T/T_v \sqcup T'/T'_w \sqcup \hat{F}$.

- External Merge and Internal Merge satisfy various cost optimization measures, while Sideward Merge is non-optimal, with a hierarchy of non-optimality among the different cases, see [19].
- when all the possible choices of syntactic objects S and S' are considered, the Merge operation can be assembled into a single $\mathcal{K} = \sum_{S, S'} \mathfrak{M}_{S, S'}$ (which despite looking like an infinite sum always results in a finite sum when applied to a workspace F as only a finite number of accessible terms are present in a given workspace)

$$(1.3) \quad \mathcal{K} = \sum_{S, S'} \mathfrak{M}_{S, S'} = \sqcup \circ (\mathfrak{B} \otimes \text{id}) \circ \Pi_{(2)} \circ \Delta,$$

where $\Pi_{(2)}$ is the projection that selects the terms with two components in the left channel of the coproduct. The map (1.3) is a Hopf algebra Markov chain.

We refer the reader to [17] for a more detailed account of this model of syntax and the Merge action. We recalled it here for comparison, to easily outline the similarities and differences with the case of morphology and because it will become a part of the overall morphosyntactic system.

2. MODELING MORPHOLOGY

As mentioned in the Introduction, our linguistic approach to the mathematical modeling of morphology relies on the ideas of Distributed Morphology, as developed by Halle and Marantz in [10] and [11]. This being said, our formulation can be seen as being also, in some respects, related to the Nanosyntax approach, in the sense that the basic structure building magma operation is common to both syntax and morphology.

In DM, features are housed in terminal nodes of the syntactic trees. There are rules which manipulate these features on trees: fusion (combining two separate feature bundles into a singular feature bundle), fission (separating one feature bundle into two distinct ones), impoverishment (removal of one or more features in the feature bundle), and obliteration (the complete elimination of a feature bundle). We will be discussing these operations explicitly in §5 below.

These features are mapped to phonological forms via vocabulary insertion rules which specify individual mappings of feature bundles to a phonological form. Finally, phonology occurs, resulting in the surface phonological forms from the phonological forms specified from the feature-to-phonology mapping, which also takes into account other factors (such as, for instance, vowel harmony). In our model phonology is incorporated at Externalization. We focus here only on the pre-Externalization part, hence the morphological objects and workspaces that we consider will have *only* morphological features at the leaves, not morphemes. This is because our morphological objects and workspaces are language-independent—vocabulary items are only inserted in Externalization, after the morphosyntactic trees have been assembled and reach their final configuration. Note that features can also be considered to be language-dependent, as one can claim that certain features are present

in some languages and not in others. However, our morphosyntactic workspace contains the set of all morphological features—if certain features are considered to be incompatible with a given language, these morphological structures/morphosyntactic trees containing those features will be properly filtered out at the Externalization stage.

The DM formulation of morphology is very suitable for this approach as it relies on the following two guiding principles: (a) *syntax all the way down*, and (b) *late insertion* (no phonology takes place until the structures are completely built). These ideas very much parallel those of [17], which takes the perspective that the machinery utilized in syntax can also be used in other places, hence minimizing the amount of computational architecture needed by reusing compositional tools already established, e.g. the utilization of syntactic tools (Merge and the coproduct) in the syntax-semantics interface. This philosophy clearly parallels (a). Secondly, with respect to (b), the mathematical model of [17] operates with a discrete, sequential approach, where all of the language-independent syntax is developed prior to the process of Externalization, and there is a clear division between the core computational process of syntax and the interfaces (Externalization at the Sensory-Motor interface and the Syntax-Semantics or Conceptual-Intentional interface). The modeling of morphology should be compatible with this overall organization of the Faculty-of-Language system.

2.1. Feature bundles and the magma of morphological objects. The main aspect of our formalization of morphology consists of replacing feature bundles as sets with feature bundles as tree structures. We do this in two successive steps. The first step consists of forming tree structures (our *morphological objects*) by combining features in a hierarchical way, obtaining non-planar binary rooted trees with leaves labelled by features and internal vertices labelled by feature bundles.

These morphological objects form a free non-associative commutative magma $\mathcal{MO}$ generated by the set $\mathcal{MO}_0$ of morphological features. All trees obtained in this way are full binary trees. We will then extend this construction in §2.2 so as to obtain also binary trees that contain non-branching vertices, as these are of use in representing bundles of features resulting from processes in the DM model of morphology. The tree structure of morphological objects is consistent with the idea of feature hierarchies, as formulated for instance in [24].

Thus, our main claim here is that we can consider bundles of features to have an inherent structure, and we can hence model a feature bundle as tree structures. The idea that features can have hierarchical relations with respect to each other has been explored through the notion of Feature Geometries [13], and modeling morphology as tree structures has been proposed at various points to varying degrees. Within nanosyntax, this is a feature of the system, given that the morphological structures are assembled via the syntax [3, 2]. Moreover, [7] and [8] utilize feature geometries to inform tree representations of English nominal inflection morphemes and tense-aspect-mood (TAM) morphemes in English and Spanish, respectively. Within DM, [12] uses unary-branching trees as the internal structure for phi-features.¹ Generalizing these ideas to make them consistent throughout all morphological features/structures, we take the view that morphological structures (feature bundles) can be modeled as hierarchical tree structures comprised of individual features at leaves that have been iteratively merged. This entails the existence of a morphism between a bundle of features and a binary tree structure.

¹For more instances of representing morphological structures as trees, see [21, 26, 27, 6, 21, 22, 29], among others.

In particular, in the model we develop here, we will not consider morphological trees with higher valence nodes. This is consistent with typical use in Distributed Morphology and with Nanosyntax. However, trees with higher valence nodes are in use in Feature Geometry. Incorporating more general forms of morphological trees is possible. It requires extending the magma we discuss here that generates our morphological tree with additional n -ary operations for higher $n \geq 3$. The use of binary trees simplifies the structure and provides greater consistency in the interface with syntax, as we will be discussing in the following sections. Thus, unless a compelling reason exists for requiring higher valences, the binary structures appear preferable. We will discuss briefly at the end of the paper what changes in the interface with syntax if higher valence morphological trees are used.

We introduce, as the basic algebraic structure for morphology, the same fundamental structure we have in the case of syntax, namely a magma operation.

Definition 2.1. *Let $\mathcal{MO}_0$ denote the (finite) set of morphological features (such as $[\pm\text{PL}]$ for the valued forms and $[\text{uPL}]$ for the unvalued form of the plural feature, etc.). Let $\mathcal{MO}$ denote the free non-associative commutative magma generated by the set $\mathcal{MO}_0$*

$$(2.1) \quad \mathcal{MO} = \text{Magma}_{na,c}(\mathcal{MO}_0, \mathfrak{M}^{\text{morph}}).$$

As in the case of syntactic objects we can identify $\mathcal{MO} \simeq \mathfrak{T}_{\mathcal{MO}_0}$ with the set of non-planar full binary rooted trees with leaves decorated by elements of $\mathcal{MO}_0$. Unlike in syntax, here we also endow trees $S \in \mathfrak{T}_{\mathcal{MO}_0}$ with a labeling of the non-leaf vertices that is completely determined by the labels at the leaves and the tree structure. The non-leaf vertices are labelled by sets in $\mathcal{P}(\mathcal{MO}_0) = 2^{\mathcal{MO}_0}$ (the power set of $\mathcal{MO}_0$) in such a way that, if the two vertices v_1, v_2 below a given vertex v are labelled by subsets B_{v_1} and B_{v_2} of $\mathcal{MO}_0$, then v is labelled by

$$(2.2) \quad B_v = B_{v_1} \cup B_{v_2}.$$

Thus, the root vertex of $S \in \mathfrak{T}_{\mathcal{MO}_0}$ is labelled by the set $\cup_{\ell \in L(S)} B_\ell$, where $L(S)$ is the set of leaves of M and $B_\ell = \{\mu_\ell\}$ is a single feature $\mu_\ell \in \mathcal{MO}_0$ associated to the leaf. We refer to the elements of $\mathcal{MO} \simeq \mathfrak{T}_{\mathcal{MO}_0}$ with this labeling as morphological objects or morphological trees, and to the sets $B \in \mathcal{P}(\mathcal{MO}_0) = 2^{\mathcal{MO}_0}$ as bundles of features or feature bundles.

The labeling of internal vertices of morphological trees shows that we can regard them as assembly procedures for bundles of morphological features. These tree structures should be thought of as templates for word formation when morphemes and vocabulary items are inserted in Externalization.

Remark 2.2. Note that we take $B_v = B_{v_1} \cup B_{v_2}$ in (2.2) rather than $B_{v_1} \sqcup B_{v_2}$. This means that, for example, a feature bundle $B_v = [\alpha, \beta, \phi]$ may be obtainable both as

$$(2.3) \quad \begin{array}{c} [\alpha, \beta, \phi] \\ \wedge \\ \alpha \quad [\beta, \phi] \\ \wedge \\ \beta \quad \phi \end{array} \quad \text{or} \quad \begin{array}{c} [\alpha, \beta, \phi] \\ \wedge \\ [\alpha, \phi] \quad [\beta, \phi] \\ \wedge \quad \wedge \\ \alpha \quad \phi \quad \beta \quad \phi \end{array}$$

as well as other possible tree configurations. This fact will be useful in §5.2 to model Distributed Morphology operations like “fission”. Note that free structure formation (in morphology as in syntax) typically overgenerates, and that other filtering mechanisms intervene to eliminate ill formed structures and tame the combinatorial explosion, such as coloring rules (like those governing theta roles and phases in syntax) and filtering by morphosyntactic parameters in Externalization.

The first tree in (2.3), and in general similar comb-like trees, represent bundles of features that belong to a single feature hierarchy. Feature hierarchies should be modeled by a partial order or a preorder. Thus, other tree topologies will occur to reflect the possibilities of features belonging to different unrelated hierarchies.

Remark 2.3. We represent features as bivalent in this work. That being said, the mathematical formulation of morphosyntax developed in this paper is compatible with the choice of either privative (the feature simply exists or does not, so e.g. the plural feature would be represented as [PL] and singular would be represented with a different feature [SG]) or bivalent (every feature can be + or -, so that plural and singular number for example can be represented by [+PL] and [-PL], respectively) features.

One mathematical argument in favor of features being bivalent is that the size of the feature space is greatly decreased when utilizing the binary(/ternary, including u for unvalued) scale in combination with the feature categories. That is, given n feature categories (number, person, etc.) and three valuations (+, - and u), for a total of $n + 3$ objects, the combinations can yield $3n$ unique features/feature valuations. On the other hand, this same number of feature valuations would require the much larger (whenever $n > 1$) set of $3n$ objects, as every feature valuation would be expressed as a different feature (plural and singular are different features instead of different valuations of the same PL feature).

We also make another assumption about the set of feature bundles and the set of lexical items and syntactic features that labels the leaves of the syntactic trees.

In addition to morphological features, hierarchies, and their assembly into morphological trees via the magma operation, we need a rule establishing the ways in which morphological feature bundles can be matched to syntactic data (lexical items and syntactic features in $\mathcal{SO}_0$ at the leaves of syntactic trees).

Definition 2.4. *There is a correspondence through which feature bundles can be matched with lexical items and syntactic features, namely a subset $\Gamma_{SM} \subset \mathcal{P}(\mathcal{MO}_0) \times \mathcal{SO}_0$ such that the second projection $\pi_2 : \mathcal{P}(\mathcal{MO}_0) \times \mathcal{SO}_0 \rightarrow \mathcal{SO}_0$ restricted to Γ_{SM} is surjective*

$$\pi_2|_{\Gamma_{SM}} : \Gamma_{SM} \twoheadrightarrow \mathcal{SO}_0.$$

We refer to Γ_{SM} as the Syntax-Morphology feature correspondence. We say that a pair (B, α) consisting of a feature bundle $B \in \mathcal{P}(\mathcal{MO}_0)$ and an element $\alpha \in \mathcal{SO}_0$ is a matching pair iff $(B, \alpha) \in \Gamma_{SM}$.

The purpose of the correspondence Γ_{SM} is to match morphological feature bundles to syntactic features and lexical items. It is too restrictive to implement this matching through a function, because it may be multivalued (the same bundle of morphological features may be compatible with more than one element in $\mathcal{SO}_0$, but also the same lexical item may occur with different combinations of morphological features). We do want, however, the surjectivity of $\pi_2|_{\Gamma_{SM}} : \Gamma_{SM} \twoheadrightarrow \mathcal{SO}_0$ since we want all elements of $\mathcal{SO}_0$ to be able to carry some morphological features. (However, see Remark 5.12 for a situation where it is preferable to drop this surjectivity condition.)

For example, a plural feature in its valued forms $[\pm\text{PL}]$ can match accompanying nouns so that a pair $([\pm\text{PL}], \mathbf{N}) \in \Gamma_{SM}$. It also exists in its unvalued form $[u\text{PL}]$, to be used in syntactic heads like $\mathbf{T}$, with $([u\text{PL}], \mathbf{T}) \in \Gamma_{SM}$, whose feature is unvalued exactly until morphological elements have been plugged into the leaves of the syntax trees to create morphosyntactic trees (as we will discuss in §3). Later in the system, Agree will target unvalued morphemes

such as this unvalued PL morpheme in a T-head, because its value is context-sensitive: it will depend on other components of the syntax tree (e.g. the subject) which cannot be evaluated at this earlier point in the derivation, within the morphological workspace before the morphosyntax trees have been constructed. (This is similar to the coloring problems for theta roles and phases analyzed in [20] and [18].) The formalization of Agree will be discussed separately from this work.

2.2. Non-branching vertices and feature bundles. In morphological tree, unlike the case of syntactic trees, it is desirable to also allow non-branching vertices. We can think of trees such as

$$\begin{array}{c} [\alpha, \beta, \phi] \\ \swarrow \quad \searrow \\ \alpha \quad [\beta, \phi] \\ \quad \quad | \\ \quad \quad \beta \end{array}$$

as representing, in the non-branching node labelled $[\beta, \phi]$, the addition of a feature ϕ that eventually modifies the realization of the feature β but does not itself carry a place to be realized as an independent morpheme in vocabulary insertion, unlike the case of a tree of the form

$$\begin{array}{c} [\alpha, \beta, \phi] \\ \swarrow \quad \searrow \\ \alpha \quad [\beta, \phi] \\ \quad \quad \swarrow \quad \searrow \\ \quad \quad \beta \quad \phi \end{array}$$

While full binary trees (with no non-branching vertices) can be generated by the magma, incorporating trees with non-branching vertices requires more than just the magma structure. Indeed, this is where morphology makes use of a coproduct structure. We show here that trees with non-braching vertices can be obtained from the morphological trees generated by the magma through the algebraic structure of *comodule over a coalgebra*.

In the modeling of syntax, the coproduct structure on the span of syntactic workspaces comes in three different flavors, denoted by Δ^c , Δ^ρ , and Δ^d in [17], with somewhat different algebraic properties (see §1.2 of [17]). In all of these forms, the left-channel of the coproduct is the same, and it contains forests of extracted accessible terms $F_{\underline{v}} = T_{v_1} \sqcup \cdots \sqcup T_{v_k}$, while the difference is in the way the remaining term of the extraction, in the right-channel of the coproduct, is obtained. In the coproduct Δ^c the quotients $T/^c F_{\underline{v}}$ are obtained by shrinking each accessible term T_{v_i} of $F_{\underline{v}}$ to its root vertex v_i that remains labelled by what will be the trace of movement. In the coproduct Δ^d the terms $T/^d F_{\underline{v}}$ are the maximal full binary tree obtained by edge contractions from the tree with non-branching vertices resulting from cutting off $F_{\underline{v}}$. It is argued in [17] that these two forms of the coproduct serve different purposes: the one that keeps the trace needed for interpretation at the syntax-semantics interface and the one that does not keep the trace representing the form at Externalization (where the trace is not externalized). In the third coproduct Δ^ρ , intermediate between these two, the terms $T/^{\rho} F_{\underline{v}}$ are non-full binary trees that contain non-branching vertices (where the cuts removing $F_{\underline{v}}$ are performed). This form of the coproduct does not directly play a role in the model of syntax and the sensory-motor (Externalization) and conceptual-intensional (syntax-semantics) interfaces.

We argue here that, instead, the form Δ^ρ of the coproduct is useful for modeling the interface between syntax and morphology. To this purpose, we first review more carefully the properties of this coproduct.

2.3. Coproduct and comodule structure. A left comodule $\mathcal{N}$ for a coalgebra $(\mathcal{C}, \Delta)$ is a vector space with a linear map $\rho_L \in \text{Hom}(\mathcal{N}, \mathcal{C} \otimes \mathcal{N})$ satisfying

$$(\text{id}_{\mathcal{C}} \otimes \rho_L) \circ \rho_L = (\Delta \otimes \text{id}_{\mathcal{N}}) \circ \rho_L \quad \text{and} \quad (\epsilon \otimes \text{id}_{\mathcal{N}}) \circ \rho_L = \text{id}_{\mathcal{N}},$$

with ϵ the coproduct and counit of $\mathcal{C}$. A right comodule is defined similarly with a linear map $\rho_R \in \text{Hom}(\mathcal{N}, \mathcal{N} \otimes \mathcal{C})$ satisfying

$$(\rho_R \otimes \text{id}_{\mathcal{C}}) \circ \rho_R = (\text{id}_{\mathcal{N}} \otimes \Delta) \circ \rho_R \quad \text{and} \quad (\text{id}_{\mathcal{N}} \otimes \epsilon) \circ \rho_R = \text{id}_{\mathcal{N}}.$$

A bicomodule $\mathcal{N}$ for a coalgebra $\mathcal{C}$ is both a left and a right comodule, with the compatibility condition expressed by the identity

$$(2.4) \quad (\text{id}_{\mathcal{C}} \otimes \rho_R) \circ \rho_L = (\rho_L \otimes \text{id}_{\mathcal{C}}) \circ \rho_R.$$

A coalgebra $\mathcal{C}$ is a bicomodule over itself with $\rho_L = \rho_R = \Delta$.

Definition 2.5. As in [17], we use the notation $\mathfrak{T}_{\mathcal{MO}_0}^{\leq 2}$ and $\mathfrak{F}_{\mathcal{MO}_0}^{\leq 2}$ for the set of binary rooted trees (respectively, forests) that can contain non-branching vertices (so all vertices have ≤ 2 descendents), with leaves decorated by elements of the set $\mathcal{MO}_0$. We denote by $\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0}^{\leq 2})$ the vector space (over $\mathbb{Q}$ or $\mathbb{R}$) spanned by the forests in $\mathfrak{F}_{\mathcal{MO}_0}^{\leq 2}$. We also write $\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0})$ for the vector space spanned by the forests in the set $\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0})$, where components are in the set $\mathcal{MO}$ of (2.1).

Remark 2.6. As shown in [17], the vector space $\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0}^{\leq 2})$ is a graded connected Hopf algebra with product and coproduct $(\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0}^{\leq 2}), \sqcup, \Delta^\rho)$. The vector space $\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0})$ is a bicomodule over the Hopf algebra $(\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0}^{\leq 2}), \sqcup, \Delta^\rho)$. The right-comodule structure is given by

$$(2.5) \quad \rho_R = \Delta^\rho : \mathcal{V}(\mathcal{F}_{\mathcal{MO}_0}) \rightarrow \mathcal{V}(\mathcal{F}_{\mathcal{MO}_0}) \otimes \mathcal{V}(\tilde{\mathcal{F}}_{\mathcal{MO}_0}^{\leq 2}).$$

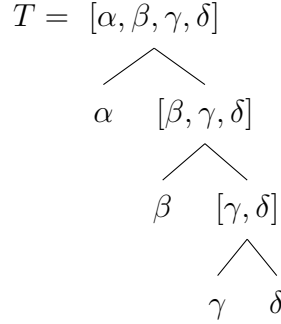
The unit 1 of the algebra $(\mathcal{V}(\mathcal{F}_{\mathcal{MO}_0}), \sqcup)$ (the formal empty forest) is also the unit of the algebra $(\mathcal{V}(\tilde{\mathcal{F}}_{\mathcal{MO}_0}^{\leq 2}), \sqcup)$, hence we obtain a left-comodule structure on $\mathcal{V}(\mathcal{F}_{\mathcal{MO}_0})$ by taking $\rho_L = 1 \otimes \text{id}$. These satisfy the compatibility (2.4).

The right-comodule structure is the interesting part here, because it describes the property that, when applying the coproduct Δ^ρ to an object $T \in \mathfrak{T}_{\mathcal{MO}_0}$ (or a workspace $F \in \mathfrak{F}_{\mathcal{MO}_0}$) the left-channel of the coproduct Δ^ρ is always also in $\mathfrak{F}_{\mathcal{MO}_0}$, while the right channel is in the larger $\mathfrak{F}_{\mathcal{MO}_0}^{\leq 2}$. In other words, because the coproduct is always applying to objects of the morphological magma, all the subtrees of the morphological tree that the coproduct applies to are also elements of that magma, and hence the coproduct can only remove binary-branching morphological trees. On the other hand, the quotient part of the coproduct can leave behind a unary-branching structure that is part of the extended morphological workspace which will be defined in the next subsection.

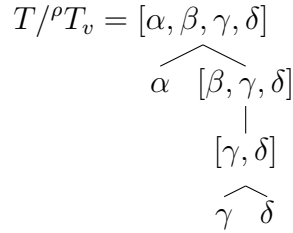
2.4. The space of morphological workspaces. A significant difference with respect to the syntactic objects is that the morphological objects in $\mathcal{MO}$ carry a labeling of the internal vertices by bundles of features $B_v \in \mathcal{P}(\mathcal{MO}_0)$ constructed on the basis of the features in $\mathcal{MO}_0$ assigned at the leaves.

The quotients $T/\rho T_v$ of morphological objects $T \in \mathcal{MO}$, which occur in the right-channel of the coproduct Δ^ρ maintain at the non-branching vertices the features that were contributed by the leaves of T_v . We can see this in the following example.

Example 2.7. Consider a morphological tree in $\mathcal{MO}$ of the form



and consider the term in the coproduct $\Delta^\rho(T)$ of the form $T_v \otimes T/\rho T_v$ where $T_v = \beta$ is the single leaf marked by the feature $\beta \in \mathcal{MO}_0$. The corresponding quotient term $T/\rho T_v$ is of the form



Note that extended morphological objects also include, for example, single nodes labelled by a bundle of morphological features instead of a single feature. These, like the quotient $T/\rho T_v$ of the previous example, are extended morphological trees that cannot be generated by the magma $(\mathcal{MO}, \mathfrak{M}^{\text{morph}})$, but they can be generated, starting from objects of the magma $\mathcal{MO}$ by applying the coproduct Δ^ρ and considering the terms that arise in the right-channel of the coproduct (that is, they are generated by the right-comodule structure of $\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0})$,

Thus, in the case of objects in $\mathfrak{T}_{\mathcal{MO}_0}^{\leq 2}$, if we want to keep track of the assignments of feature bundles B_v at the internal vertices, in such a way that the right-comodule structure still works, we need to allow for more general assignments than those determined by the features at the leaves.

Definition 2.8. The extended morphological objects are pairs (T, B) of a tree $T \in \mathfrak{T}_{\mathcal{MO}_0}^{\leq 2}$ and an assignment $B : V^o(T) \rightarrow \mathcal{P}(\mathcal{MO}_0)$ with the properties:

- For $v, w \in V^o(T)$ with $T_w \subset T_v$ the feature bundles satisfy $B_w \subseteq B_v$.
- For all $v \in V^o(T)$, the feature bundle $B_v \in \mathcal{P}(\mathcal{MO}_0)$ contains the set $\cup_{\ell \in L(T_v)} \{\mu_\ell\}$ of all the morphological features $\mu_\ell \in \mathcal{MO}_0$ assigned to the leaves $\ell \in L(T_v)$.
- If $T_v \subseteq T$ does not contain any non-branching vertices, then $B_v = \cup_{\ell \in L(T_v)} \{\mu_\ell\}$, with $\mu_\ell \in \mathcal{MO}_0$ the morphological features at the leaves.

The morphological workspaces are forests $F = \sqcup_a T_a$ where all the components T_a are extended morphological objects. We use the notation

$$(2.6) \quad \widetilde{\mathcal{MO}} := \mathfrak{T}_{\mathcal{MO}_0, \mathfrak{B}}^{\leq 2} \quad \text{and} \quad \mathcal{W}_{\mathcal{M}} := \mathfrak{F}(\widetilde{\mathcal{MO}})$$

for the set $\widetilde{\mathcal{MO}}$ of extended morphological objects, where $\mathfrak{T}_{\mathcal{MO}_0, \mathfrak{B}}^{\leq 2}$ means the set of pairs (T, B) as above, and the set $\mathcal{W}_{\mathcal{M}}$ of morphological workspaces, where $\mathfrak{F}(\widetilde{\mathcal{MO}})$ means the set of forests whose components are trees in $\mathfrak{T}_{\mathcal{MO}_0, \mathfrak{B}}^{\leq 2}$. We write $\mathcal{V}(\mathcal{W}_{\mathcal{M}})$ for the vector space spanned by the set of morphological workspaces.

Thus, the difference between the morphological trees and the extended morphological trees is that the latter have non-branching vertices and the bundles of features labeling internal vertices can contain additional features that are not contained in the set of features assigned at the leaves. We will discuss in §5 why this is necessary to properly represent Distributed Morphology.

The assignment of feature bundles B_v to extended morphological objects, with the rules of Definition 2.8 ensures that the following holds.

Lemma 2.9. *The vector space $\mathcal{V}(\mathcal{W}_{\mathcal{M}})$ of morphological workspaces is a graded connected Hopf algebra with product $\sqcup$ and coproduct Δ^ρ and the vector subspace $\mathcal{V}(\mathfrak{F}_{\mathcal{MO}_0}) \subset \mathcal{V}(\mathcal{W}_{\mathcal{M}})$ is a right-comodule as in (2.5).*

We refer to $(\mathcal{V}(\mathcal{W}_{\mathcal{M}}), \sqcup, \Delta^\rho)$ as the Hopf algebra of morphological workspaces.

In syntax, the coproduct structure is needed for the extraction of accessible terms (and corresponding cancellation of deeper copies) that are needed for movement. External Merge, by itself, could otherwise be accounted for already by the magma structure of syntactic objects. Workspaces and the coproduct make it possible to unify External Merge and Internal Merge into a single operation definable in Hopf algebra terms. In morphology one does not need movement, so in principle the magma operation would suffice for structure building, except for allowing for non-branching vertices, which as we discussed, can be obtained using the comodule $\rho_R = \Delta^\rho$. Note however that, even though we do use a coproduct/comodule structure, we do not need to introduce in morphology a Merge-like operation: any morphological tree that looks like $\mathfrak{M}(T, T')$ where T, T' are either in $\mathcal{MO}$ or in $\widetilde{\mathcal{MO}}$, is already present in either the magma $\mathcal{MO}$ or in the terms in the right-channel of the coproduct Δ^ρ applied to elements of the magma $\mathcal{MO}$.

Thus, unlike the case of syntax, we do not need to introduce a Merge action on morphological workspaces, rather we will have a different kind of structure formation operation, which still relies on the Hopf algebra structure, and takes care of interfacing syntax with morphology, leading to the creation of morphosyntactic objects. We describe this mechanism in the next sections §3 and §4.

The points of view presented in §3 and §4 describe the formation of morphosyntax from syntactic and morphological data with two slightly different perspectives. In §3 we focus on the morphological data inserted at the leaves of syntactic trees, hence the algebraic formalism revolves around operads and algebras over operads. In §4 we focus on morphological workspaces and operations that use syntactic objects and that collect inputs from morphological workspaces to produce morphosyntax. These two viewpoints are equivalent in terms of the resulting structure formation. It is useful to develop both perspective for the same reason discussed in [20] and [18]: in view of developing a model of Agreement, it is important to be able to formulate filtering of structures via the formalism of (colored) operads and algebras over operads, while this also need to be implementable alongside structure formation, via a description in terms of maps acting on workspaces. The results of §3 and §4 build the necessary theoretical setting.

3. MORPHOSYNTAX AND ALGEBRAS OVER OPERADS

The main idea, as in various existing models of morphology, is that morphological features exist as structure at the leaves of syntactic trees. An analogy from a physicist's perspective would be that the data $\alpha \in \mathcal{SO}_0$ of lexical items and syntactic features at the leaves of syntactic objects acquire inner structure (further degrees of freedom) when one zooms in at the scale of morphology (word formation) rather than at the large scale structure of syntax (sentence formation). This means, in terms of mathematical formulation, that we are looking at an operation that inserts structure at the leaves of a tree: this naturally suggests that the formalism of operads and algebras over operads is the right type of algebraic structure to describe this kind of model. However, as we will see, morphological structures do not directly form an algebra over an operad themselves, but the syntactic objects, with their structure of an algebra over an operad, determine structure formation operations on the morphological workspaces that combine syntactic objects and morphological trees forming the morphosyntactic structures. These morphosyntactic objects, in turn, form another algebra over the same operad.

3.1. Operads and algebras over operads. The notion of operad describes the compositions of operations with multiple inputs and a single output, where the output of one operation can serve as input of another one.

An operad (in the category of sets) is a collection $\mathfrak{O} = \{\mathfrak{O}(n)\}$ of sets $\mathfrak{O}(n)$ whose elements are operations $T \in \mathfrak{O}(n)$ that have n inputs and one output. The algebraic structure governing this collection of sets $\mathfrak{O}$ consists of *compositions* that relate the operations in the sets $\mathfrak{O}(n)$. These compositions are usually presented in two different forms, one that saturates all the inputs of an operation with outputs of other operations, and one where only one input at a time is filled with one output. These two different formulations of the operad structure are equivalent (for unital operads). The formulation with simultaneous saturation of all inputs is presented as compositions of the form

$$(3.1) \quad \gamma : \mathfrak{O}(n) \times \mathfrak{O}(k_1) \times \cdots \times \mathfrak{O}(k_n) \rightarrow \mathfrak{O}(k_1 + \cdots + k_n)$$

where γ takes the single output of an operation in $\mathfrak{O}(k_j)$ and feeds it into the j -th input of an operation in $\mathfrak{O}(n)$. Since this is done for every input of operations in $\mathfrak{O}(n)$ the result is now an operation that only has the inputs coming from the inputs of the operations in $\mathfrak{O}(k_j)$ (a total of $k_1 + \cdots + k_n$ inputs), and a single output, and the composition rule (3.1) requires that the operation obtained in this way is in the set $\mathfrak{O}(k_1 + \cdots + k_n)$. The composition operations (3.1) are also required to satisfy an associativity rule, namely

$$(3.2) \quad \begin{aligned} \gamma(\gamma(T, T_1, \dots, T_n); T_{1,1}, \dots, T_{1,m_1}, \dots, T_{n,1}, \dots, T_{n,m_n}) = \\ \gamma(T; \gamma(T_1; T_{1,1}, \dots, T_{1,m_1}), \dots, \gamma(T_n; T_{n,1}, \dots, T_{n,m_n})). \end{aligned}$$

The second form of the operad composition, with a single output-input match, has composition rules (operad insertions) of the form

$$(3.3) \quad \circ_i : \mathfrak{O}(n) \otimes \mathfrak{O}(m) \rightarrow \mathfrak{O}(n + m - 1),$$

satisfying

$$(X \circ_j Y) \circ_i Z = \begin{cases} (X \circ_i Z) \circ_{j+c-1} Y & 1 \leq i < j \\ X \circ_j (Y \circ_{i-j+1} Z) & j \leq i < b+j \\ (X \circ_{i-b+1} Z) \circ_j Y & j+b \leq i \leq a+b-1. \end{cases}$$

for $1 \leq j \leq a$ and $b, c \geq 0$, with $X \in \mathfrak{D}(a)$, $Y \in \mathfrak{D}(b)$, and $Z \in \mathfrak{D}(c)$. When the operad is unital, namely if there is an operation (unit) $\mathbf{1} \in \mathfrak{D}(1)$ that satisfies $\gamma(\mathbf{1}; T) = T$ and $\gamma(T; \mathbf{1}, \dots, \mathbf{1}) = T$, the composition operations (3.1) are equivalent to the insertion operations (3.3), with the laws γ of (3.1) obtained from the insertions $\circ_i$ of (3.3) as

$$(3.4) \quad \gamma(X, Y_1, \dots, Y_n) = (\dots (X \circ_n Y_n) \circ_{n-1} Y_{n-1}) \dots \circ_1 Y_1).$$

In addition to these properties, operads can have a symmetric property that makes composition rules compatible with permutations of the inputs, but we will not be discussing them here.

The notion of algebra over an operad describes sets whose elements can serve as inputs and outputs of the operations in the operad.

An algebra A over an operad $\mathfrak{D}$ (in the category of sets) is a set A with an action of the operad $\mathfrak{D}$, namely operations

$$(3.5) \quad \gamma_A : \mathfrak{D}(n) \times A^n \rightarrow A$$

satisfying

$$(3.6) \quad \begin{aligned} \gamma_A(\gamma(T; T_1, \dots, T_n); a_{1,1}, \dots, a_{1,k_1}, \dots, a_{n,1}, \dots, a_{n,k_n}) = \\ \gamma_A(T; \gamma_A(T_1; a_{1,1}, \dots, a_{1,k_1}), \dots, \gamma_A(T_n; a_{n,1}, \dots, a_{n,k_n})) \end{aligned}$$

One interprets here γ_A as the operation that takes n inputs a_i from the set A (an element $\underline{a} = (a_i)_{i=1}^n$ in the set A^n) and inserts them in the inputs of an n -ary operation $T \in \mathfrak{D}(n)$, which then produces as single output another element of the same set A .

The condition (3.5) means that one can compose operations according to the composition rules γ in the operad $\mathfrak{D}$ and then apply them to inputs in A , or one can *equivalently* apply the first operations in $\mathfrak{D}$ to inputs in A using the rule γ_A and then insert the resulting outputs (also in A) as inputs to the second operation in $\mathfrak{D}$, again according to the rule γ_A .

We also recall the notion of a colored operad, which is a collection $\mathfrak{D} = \{\mathfrak{D}(c, c_1, \dots, c_n)\}$ of sets, with $c, c_i \in \Omega$ for $i = 1, \dots, n$, where Ω is a (finite) set of color labels. The c_i are the colors assigned to the inputs of the n -ary operations in $\mathfrak{D}(c, c_1, \dots, c_n)$, and c is the color label assigned to the output. The composition is then like the usual operad composition but with the requirement that colors should match, namely

$$(3.7) \quad \begin{aligned} \gamma : \mathfrak{D}(c, c_1, \dots, c_n) \times \mathfrak{D}(c_1, c_{1,1}, \dots, c_{1,k_1}) \times \dots \times \mathfrak{D}(c_n, c_{n,1}, \dots, c_{n,k_n}) \\ \rightarrow \mathfrak{D}(c, c_{1,1}, \dots, c_{1,k_1}, \dots, c_{n,1}, \dots, c_{n,k_n}). \end{aligned}$$

These composition operations satisfy the same associativity and unit conditions as in the usual ones of the non-colored case. Instead of a single unit there is now a unit $\mathbf{1}_c \in \mathfrak{D}(c, c)$ for each color $c \in \Omega$.

An algebra over a colored operad is a collection of sets $A = \{A^{(c)}\}_{c \in \Omega}$ with the operad action (satisfying the same compatibility condition with the compositions γ as in the non-colored case) of the form

$$\gamma_A : \mathfrak{D}(c, c_1, \dots, c_n) \times A^{(c_1)} \times \dots \times A^{(c_n)} \rightarrow A^{(c)}.$$

3.2. Syntactic objects as an algebra over an operad. Before we can discuss how syntactic objects and morphological objects interface it is helpful to recall some results from [17] (see also [20] and [18]) about realizing syntactic objects as an algebra over an operad.

Operads and algebras over operads have already been used, in modeling syntax, in [17]. We recall here how to view syntactic objects as an algebra over an operad, as that will be the basis for introducing the further structure needed to generate morphosyntactic trees.

The *Merge operad* $\mathcal{M}$ has $\mathcal{M}(n)$ given by the set of all abstract (non-planar) binary rooted trees with n (non-labelled) leaves. The insertion operations $T \circ_\ell T'$, for $\ell \in L(T)$ graft the root vertex of $T' \in \mathcal{M}(m)$ to the leaf ℓ of $T \in \mathcal{M}(n)$, resulting in a tree $T \circ_\ell T'$ in $\mathcal{M}(n + m - 1)$. The set $\mathcal{SO} = \mathfrak{T}_{\mathcal{SO}_0}$ of syntactic objects is then an algebra over the operad $\mathcal{M}$ with the operad action $\gamma_{\mathcal{SO}} : \mathcal{M}(n) \times \mathfrak{T}_{\mathcal{SO}_0}^n \rightarrow \mathfrak{T}_{\mathcal{SO}_0}$ that plugs the root of the syntactic object $T_\ell \in \mathfrak{T}_{\mathcal{SO}_0}$ to the ℓ -th leaf of the tree $T \in \mathcal{M}(n)$.

We will equivalently use the notation $\gamma(T, T_1, \dots, T_n)$ or $\gamma(T, \{T_\ell\}_{\ell \in L(T)})$ for these and other operad insertion operations. With the first notation we do not necessarily mean that the leaves are ordered (the trees are all non-planar), rather that they are labelled so that we know which of the n trees is matched to which leaf, as the second notation clarifies more explicitly. This is discussed also in §3.8.1 of [17].

It will be useful in the following (see §5.2) to also write the operad action $\gamma_{\mathcal{SO}}$ as a composition of individual insertions at each of the leaves, as one does in (3.4) for the operad composition. In the case of the operad composition γ , the single insertions (3.3) map $\circ_\ell : \mathcal{M}(n) \times \mathcal{M}(m) \rightarrow \mathcal{M}(n + m - 1)$. However, in the case of operad action $\gamma_{\mathcal{SO}}$, if we perform a single insertion of a syntactic object $T' \in \mathcal{SO}$ with m leaves (each of which is labeled by an element of $\mathcal{SO}_0$) at one of the leaves of a $T \in \mathcal{M}(n)$ the result will be an operation with only $n - 1$ inputs, since the m leaves of T' are not open inputs, being already filled with elements in $\mathcal{SO}_0$. So we have

$$(3.8) \quad \circ_{\mathcal{SO}_\ell} : \mathcal{M}(n) \times \mathcal{SO}_m \rightarrow \mathcal{M}_{\mathcal{SO}}(n - 1, m),$$

where we write $\mathcal{SO}_m \subset \mathcal{SO}$ for the set of syntactic objects with $m \geq 1$ leaves and we use the notation $\mathcal{M}_{\mathcal{SO}}(n - 1, m)$ to indicate the set of non-planar full binary trees on $n + m - 1$ leaves where $n - 1$ of the leaves are unlabelled and m are labelled by elements in $\mathcal{SO}_0$.

The set $\mathcal{SO}$ of syntactic objects is not itself an operad, because only the leaves and not the root of the trees in $\mathcal{SO}$ are labeled, so elements of $\mathcal{SO}$ can be acted upon by elements of $\mathcal{M}$, but not by other elements of $\mathcal{SO}$.

This leads to an additional remark, which we will not discuss in depth in this paper, but that may play a useful role in further developments.

Remark 3.1. When elements of $\mathcal{SO}$ are endowed with a head function, they have a labeling algorithm which induces a labeling of all the vertices, including the root.

Let $\text{Dom}(h) \subset \mathcal{SO}$ denote syntactic objects in the domain of a head function $h : T \mapsto h_T$. (For the definition and properties of head functions see §1.13.3 of [17].) We write the elements of $\text{Dom}(h)$ as pairs (T, h_T) .

Lemma 3.2. *The set $\text{Dom}(h) \subset \mathcal{SO}$ is a colored operad, with color set $\Omega = \mathcal{SO}_0$.*

Proof. As discussed in §1.15 of [17] for a syntactic object (T, h_T) with a head function, there is a labeling algorithm that assigns to each internal (non-leaf) vertex v of T a label $\alpha_v \in \mathcal{SO}_0$ given by the label $\alpha_v = \alpha_{h_T(v)}$ at the leaf $h_T(v)$. In particular, the root vertex v_0 of T also carries a label $\alpha_{v_0} = h_T(v_0)$, the label at the head of the entire structure T . It is then possible to insert the root vertex of a syntactic object at a leaf vertex of another one as long as their respective labels match. This gives $\text{Dom}(h) \subset \mathcal{SO}$ the structure of a colored operad. $\square$

A more sophisticated colored operad that also uses the head function on syntactic objects is introduced in [18] to account for the structure of phases.

Remark 3.3. It is important to stress that the operad structures are not a model of structure formation (which in the case of syntax is given by the Merge action on workspaces), rather a model for filtering formed syntactic objects according to coloring conditions imposed on an underlying non-colored operad and algebras over this operad, as in [18], [20].

The operad structure of Lemma 3.2 will become relevant when discussing the interaction of morphology with the coloring algorithms for syntactic objects that test for theta roles and phases, as in [20], [18], especially in view of formulating a model for agreement, but we will not discuss this further in the present paper.

3.3. Morphosyntactic trees. The operation of forming morphosyntactic trees consists of operad insertions of roots of extended morphological objects $S \in \widetilde{\mathcal{MO}}$ to the leaves of syntactic objects $T \in \mathcal{SO}$ with a matching rule Γ_{SM} (as in Definition 2.4) between the feature bundle $B_v \in \mathcal{P}(\mathcal{MO})$ at the root of S and the datum $\alpha \in \mathcal{SO}_0$ at the leaf of T where insertion is performed.

Definition 3.4. We write $\mathcal{SO}_n \subset \mathcal{SO}$ for the subset of syntactic objects with n leaves, and $\mathcal{SO}_{\{\alpha_\ell\}_{\ell \in L}} \subset \mathcal{SO}$ for the subset of syntactic objects $T \in \mathcal{SO}$ with set of leaves $L(T) = L$ and with data $\alpha_\ell \in \mathcal{SO}_0$ at the leaves $\ell \in L$, so that

$$(3.9) \quad \mathcal{SO}_n = \sqcup_{\#L=n} \mathcal{SO}_{\{\alpha_\ell\}_{\ell \in L}}.$$

Let $\widetilde{\mathcal{MO}}_B \subset \widetilde{\mathcal{MO}}$ denote the set of extended morphological objects with root vertex decorated by the feature bundle $B \in \mathcal{P}(\mathcal{MO}_0)$. The set $\mathcal{MS}$ of morphosyntactic trees is the range

$$\mathcal{MS} = \bigcup_n \gamma_{\mathcal{SO}, \mathcal{MO}}(\mathcal{SO}_n \times \widetilde{\mathcal{MO}}^n)$$

of the maps

$$(3.10) \quad \gamma_{\mathcal{SO}, \mathcal{MO}} : \mathcal{SO}_n \times \widetilde{\mathcal{MO}}^n \rightarrow \mathcal{MS}$$

with domains

$$(3.11) \quad \text{Dom}(\gamma_{\mathcal{SO}, \mathcal{MO}}) = \bigcup_n \bigcup_{\#L=n} \left\{ (T, S_1, \dots, S_n) \in \left(\mathcal{SO}_{\{\alpha_\ell\}_{\ell \in L}} \times \prod_{\ell \in L} \widetilde{\mathcal{MO}}_{B_\ell} \right) \mid (B_\ell, \alpha_\ell) \in \Gamma_{SM} \right\}$$

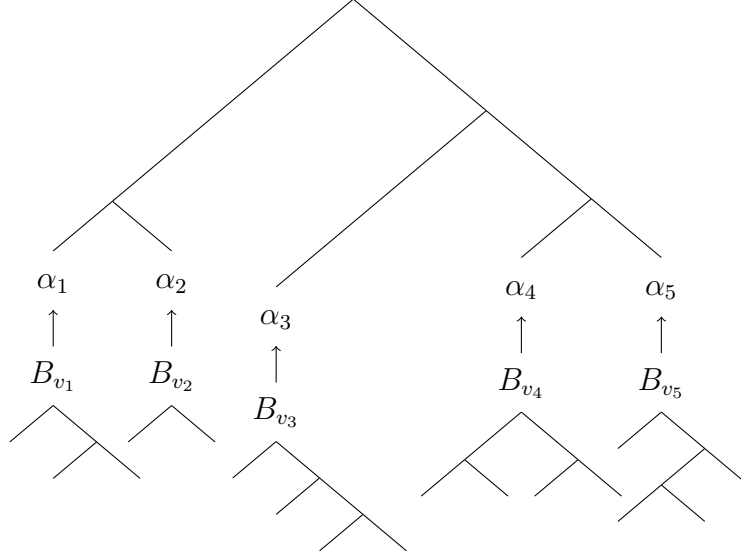
where $\Gamma_{SM} \subset \mathcal{P}(\mathcal{MO}_0) \times \mathcal{SO}_0$ is the syntax-morphology feature correspondence of Definition 2.4.

To illustrate the maps $\gamma_{\mathcal{SO}, \mathcal{MO}}$ of (3.10) consider the following example.

Example 3.5. For example, the insertion map $\gamma_{\mathcal{SO}, \mathcal{MO}} : \mathcal{SO}_5 \times \widetilde{\mathcal{MO}}^5 \rightarrow \mathcal{MS}$ performs the insertions of the roots $v_1, \dots, v_5$ of the extended morphological objects $S_1, \dots, S_5$ in the

leaves of the syntactic tree $T \in \mathcal{SO}_5$ as follows:

(3.12)



provided that the morphological feature bundles $B_{v_1}, \dots, B_{v_5}$ at the roots of these morphological trees and the lexical items and syntactic features $\alpha_1, \dots, \alpha_5$ at the leaves of the syntactic tree T satisfy the relation $(B_{v_\ell}, \alpha_\ell) \in \Gamma_{SM}$ for $\ell = 1, \dots, 5$. The morphosyntactic tree (3.12) can be written in the form of an insertion

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S_1, \dots, S_5)$$

with

$$T = \begin{array}{c} \diagup \quad \diagdown \\ \alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4 \quad \alpha_5 \end{array} \in \mathcal{SO}$$

and with

$$\begin{array}{lll} S_1 = \begin{array}{c} B_{v_1} \\ \diagup \quad \diagdown \\ \phi_{1,1} \quad \phi_{1,2} \quad \phi_{1,3} \end{array} & S_2 = \begin{array}{c} B_{v_2} \\ \diagup \quad \diagdown \\ \phi_{2,1} \quad \phi_{2,2} \end{array} & S_3 = \begin{array}{c} B_{v_3} \\ \diagup \quad \diagdown \\ \phi_{3,1} \quad \phi_{3,2} \quad \phi_{3,3} \quad \phi_{3,4} \end{array} \\ S_4 = \begin{array}{c} B_{v_4} \\ \diagup \quad \diagdown \\ \phi_{4,1} \quad \phi_{4,2} \quad \phi_{4,3} \quad \phi_{4,4} \end{array} & S_5 = \begin{array}{c} B_{v_5} \\ \diagup \quad \diagdown \\ \phi_{5,1} \quad \phi_{5,2} \quad \phi_{5,3} \quad \phi_{5,4} \end{array} & \in \widetilde{\mathcal{MO}} \end{array}$$

Like syntactic objects, the resulting morphosyntactic trees also form an algebra over an operad.

Lemma 3.6. *The set $\mathcal{MS}$ is an algebra over the Merge operad with*

$$\gamma_{\mathcal{MS}} : \mathcal{M}(n) \times \mathcal{MS}^n \rightarrow \mathcal{MS}$$

that grafts the root vertices of the n morphosyntactic trees to the unlabelled leaves of the trees in $\mathcal{M}(n)$.

Proof. The operad action $\gamma_{\mathcal{MS}}$ of the Merge operad $\mathcal{M}$ on morphosyntactic trees is directly induced by the operad action $\gamma_{\mathcal{SO}}$ on syntactic objects, since the root vertex of a morphosyntactic tree is the root vertex of a syntactic object. $\square$

Remark 3.7. In a similar way, if we consider $\text{Dom}(h) \subset \mathcal{SO}$ with the colored operad structure of Lemma 3.2, the subset $\mathcal{MS}_h \subset \mathcal{MS}$ of morphosyntactic trees of the form

$$\mathcal{MS}_h = \{\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S_1, \dots, S_n) \mid T \in \text{Dom}(h) \text{ and } S_i \in \widetilde{\mathcal{MO}}\}$$

is an algebra over the colored operad $\text{Dom}(h)$.

3.4. Morphological trees. We have seen that both $\mathcal{SO}$ and $\mathcal{MS}$ are algebras over the Merge operad $\mathcal{M}$. There is, instead, an important difference in the case of the sets $\mathcal{MO}$ and $\widetilde{\mathcal{MO}}$.

Lemma 3.8. *The sets $\mathcal{MO}$ and $\widetilde{\mathcal{MO}}$ of morphological and extended morphological trees are not algebras over the Merge operad $\mathcal{M}$.*

Proof. Operations in $\mathcal{M}$ are full binary trees with unlabeled leaves. An action of $\mathcal{M}$ on the set $\mathcal{MO}$ or $\widetilde{\mathcal{MO}}$ would require using morphological trees as inputs for elements of $\mathcal{M}$, but the labels B_v at the root of the morphological tree require syntactic information to combine with that the syntactic objects in $\mathcal{SO}$ have, in the form of elements $\alpha \in \mathcal{SO}_0$ at the leaves, but that operations in $\mathcal{M}$ do not have. We can think of this requirement, in mathematical terms as a coloring of the root of the morphological tree that requires a matching coloring of leaves for an insertion to take place, but there is no coloring of the leaves of elements $\mathcal{M}$. $\square$

In the same way, while trees in $\mathcal{MO}$ or $\widetilde{\mathcal{MO}}$ can be inserted at the leaves of trees in $\mathcal{SO}$, this again does not give these sets the structure of algebras over an operad, in the case of $\text{Dom}(h) \subset \mathcal{SO}$ with its colored operad structure of Lemma 3.2, since the result of such insertions would be morphosyntactic and not morphological trees.

3.5. Correspondences of algebras over operads. We have seen that both the set $\mathcal{SO}$ of syntactic objects and the set $\mathcal{MS}$ of morphosyntactic objects are algebras over the same operad $\mathcal{M}$. It is then natural to ask what is the relation between these two algebras-over-operads. The question of how to properly formulate this relation is relevant because the structure of algebra over an operad for syntactic objects is important to describe filtering of freely formed structures produced by Merge, especially filtering for theta role assignments, as in [20], and filtering for well-formed phases, as in [18]. These filtering are formulated in terms of coloring rules on these operads and algebras over operads. Morphosyntactic trees will also have to undergo similar filtering procedures in terms of colored operads, subject to compatibility relations with the corresponding structures on syntax. Most importantly, we expect such compatibilities of colored operads and algebras over operads to play an important role in the modeling of Agreement. Thus, it is important to provide a good formulation of the relation between these two algebras, $\mathcal{SO}$ and $\mathcal{MS}$, over the Merge operad $\mathcal{M}$.

The usual way in which one compares algebras over operads is through the following notion of morphisms.

Definition 3.9. A morphism $\varphi : A \rightarrow B$ of algebras over an operad $\mathfrak{O}$ is a map of sets satisfying the commutative diagram

$$(3.13) \quad \begin{array}{ccc} \mathfrak{O}(n) \times A^n & \xrightarrow{\gamma_A} & A \\ \text{id} \times \varphi^n \downarrow & & \downarrow \varphi \\ \mathfrak{O}(n) \times B^n & \xrightarrow{\gamma_B} & B \end{array}$$

There is, in this sense, a relation between $\mathcal{SO}$ and $\mathcal{MS}$, which is simply given by the “forgetful morphism” $\varphi : \mathcal{MS} \rightarrow \mathcal{SO}$ from morphosyntactic trees to syntactic trees that forgets the morphology by shrinking the morphological subtrees S_ℓ to their root vertex $\ell \in L(T)$, dropping the B_ℓ part of the label (B_ℓ, α_ℓ) at this vertex, resulting in just the syntactic tree:

$$\varphi : \gamma_{\mathcal{SO}, \mathcal{MO}}(T, \{S_\ell\}_{\ell \in L(T)}) \mapsto T.$$

This clearly satisfies (3.13).

However, there is another more interesting relation that follows the process of structure formation and that involves the insertion of morphology at the leaves of syntactic trees, rather than the removal of morphology from morphosyntactic trees.

Example 3.10. For example, consider again the case of the morphosyntactic tree of (3.12). We can obtain it by first applying the operad action to syntactic trees and then inserting morphological trees in the resulting syntactic tree, namely first forming

$$\gamma_{\mathcal{SO}, \mathcal{MO}} : \mathcal{M}(3) \times \mathcal{SO}_2^3 \rightarrow \mathcal{SO}, \quad T = \gamma_{\mathcal{SO}}(T', T'_1, \dots, T'_3)$$

with

$$T' = \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \end{array} \in \mathcal{M}(3)$$

where $\bullet$ marks the an inputs of operations in the operad $\mathcal{M}$, and

$$T'_1 = \alpha_1 \frown \alpha_2 \quad T'_2 = \alpha_3 \quad T'_3 = \alpha_4 \frown \alpha_5 \in \mathcal{SO}$$

and then inserting the morphological trees $S_1, \dots, S_5$ at the leaves of T with the operation

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S_1, \dots, S_5)$$

or by first inserting morphological trees in the syntactic trees, and then applying the operad action to the resulting morphosyntactic trees. This means that we can first insert the morphological trees by forming

$$\begin{aligned} \gamma_{\mathcal{SO}, \mathcal{MO}}(T'_1, S_1, S_2) &= \begin{array}{c} \diagup \quad \diagdown \\ (B_{v_1}, \alpha_1) \quad (B_{v_2}, \alpha_2) \\ \diagdown \quad \diagup \quad \diagdown \quad \diagup \quad \diagdown \quad \diagup \\ \phi_{1,1} \quad \phi_{1,2} \quad \phi_{1,3} \quad \phi_{2,1} \quad \phi_{2,2} \end{array} & \gamma_{\mathcal{SO}, \mathcal{MO}}(T'_2, S_3) &= \begin{array}{c} \diagup \quad \diagdown \\ (B_{v_3}, \alpha_3) \\ \diagdown \quad \diagup \quad \diagdown \quad \diagup \\ \phi_{3,1} \quad \phi_{3,2} \quad \phi_{3,3} \quad \phi_{3,4} \end{array} \\ \gamma_{\mathcal{SO}, \mathcal{MO}}(T'_3, S_4, S_5) &= \begin{array}{c} \diagup \quad \diagdown \\ (B_{v_4}, \alpha_4) \quad (B_{v_5}, \alpha_5) \\ \diagdown \quad \diagup \quad \diagdown \quad \diagup \quad \diagdown \quad \diagup \\ \phi_{4,1} \quad \phi_{4,2} \quad \phi_{4,3} \quad \phi_{4,4} \quad \phi_{5,1} \quad \phi_{5,2} \quad \phi_{5,3} \quad \phi_{5,4} \end{array} \end{aligned}$$

and then acting with the operad, $\gamma_{\mathcal{SO}, \mathcal{MO}} : \mathcal{M}(3) \times \mathcal{MS}_2^3 \rightarrow \mathcal{MS}$ to obtain

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T', \gamma_{\mathcal{SO}, \mathcal{MO}}(T'_1, S_1, S_2), \gamma_{\mathcal{SO}, \mathcal{MO}}(T'_2, S_3), \gamma_{\mathcal{SO}, \mathcal{MO}}(T'_3, S_4, S_5)).$$

Again, as pointed out in Remark 3.3, these operadic structures are not in themselves a model of structure formation: we will come to that more explicitly in §4 and we will discuss how the insertion operation $\gamma_{\mathcal{SO}, \mathcal{MO}}$ is involved. The analysis of these algebras over operads and their relation is discussed here as preliminary to a theory of filtering of morphosyntactic structures analogous to the filtering of syntactic structures described in [20] and [18].

Remark 3.11. Note that, in the case we are considering, not only the operad is graded by the number of inputs, $\mathcal{M} = \sqcup_{n \geq 1} \mathcal{M}(n)$, but both the algebras $\mathcal{SO}$ and $\mathcal{MS}$ over this operad also have a grading $\mathcal{SO} = \sqcup_{n \geq 1} \mathcal{SO}_n$ and $\mathcal{MS} = \sqcup_{n \geq 1} \mathcal{MS}_n$, where $\mathcal{SO}_n$ and $\mathcal{MS}_n$ denote the set of syntactic (respectively, morphosyntactic) trees with n leaves.

We also have a grading by number of leaves $\widetilde{\mathcal{MO}} = \sqcup_n \widetilde{\mathcal{MO}}_n$ on extended morphological trees. However, as discussed in Lemma 3.8, this set is not an algebra over the Merge operad.

A first step to describe more explicitly the relation between the two algebras $\mathcal{SO}$ and $\mathcal{MS}$ over the Merge operad illustrated in Example 3.10, it is convenient to first refine the notion of algebra over an operad to a graded version, which we define in the following way.

Definition 3.12. Let $\mathfrak{D}$ be an operad and A an algebra over an operad, in the category of sets. We say that A is a graded algebra over the operad $\mathfrak{D}$ if $A = \sqcup_n A_n$ for $n \geq 1$ and the operad action maps $\gamma_A : \mathfrak{D}(n) \times A^n \rightarrow A$ satisfy

$$(3.14) \quad \gamma_A : \mathfrak{D}(n) \times A_{k_1} \times \cdots \times A_{k_n} \rightarrow A_{k_1 + \cdots + k_n},$$

so that (3.6) takes the form of the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\gamma \times \text{id}} & Y \\ \downarrow \sigma & & \searrow \gamma_A \\ & & Z \\ \downarrow \text{id} \times \gamma_A^n & & \nearrow \gamma_A \\ U & \xrightarrow{\text{id} \times \gamma_A^n} & V \end{array}$$

with σ the permutation of the factors and

$$X = \mathfrak{D}(n) \times \mathfrak{D}(k_1) \times \cdots \times \mathfrak{D}(k_n) \times A_{\ell_{1,1}} \times \cdots \times A_{\ell_{1,k_1}} \times \cdots \times A_{\ell_{n,1}} \times \cdots \times A_{\ell_{n,k_n}},$$

$$Y = \mathfrak{D}(k_1 + \cdots + k_n) \times A_{\ell_{1,1}} \times \cdots \times A_{\ell_{n,k_n}},$$

$$U = \mathfrak{D}(n) \times \mathfrak{D}(k_1) \times A_{\ell_{1,1}} \times \cdots \times A_{\ell_{1,k_1}} \times \cdots \times \mathfrak{D}(k_n) \times A_{\ell_{n,1}} \times \cdots \times A_{\ell_{n,k_n}},$$

$$V = \mathfrak{D}(n) \times A_{\ell_{1,1} + \cdots + \ell_{1,k_1}} \times \cdots \times A_{\ell_{n,1} + \cdots + \ell_{n,k_n}},$$

$$Z = A_{\ell_{1,1} + \cdots + \ell_{n,k_n}}.$$

Given that both $\mathcal{SO}$ and $\mathcal{MS}$ are graded algebras over the operad $\mathcal{M}$, as in Definition 3.12, we would like to formulate their relation in a way that this graded structure is taken into account.

There is a straightforward way of extending the usual notion of morphism of algebras over operads as in Definition 3.9 to the graded case in the following way.

Definition 3.13. Similarly, a morphism of graded algebras over an operad $\mathfrak{D}$ is a collection of maps $\varphi_n : A_n \rightarrow B_n$ satisfying the commutative diagrams, for all $n \geq 1$

$$\begin{array}{ccc} \mathfrak{D}(n) \times A_{k_1} \times \cdots \times A_{k_n} & \xrightarrow{\gamma_A} & A_{k_1+\cdots+k_n} \\ \text{id} \times \varphi_{k_1} \times \cdots \times \varphi_{k_n} \downarrow & & \downarrow \varphi_{k_1+\cdots+k_n} \\ \mathfrak{D}(n) \times B_{k_1} \times \cdots \times B_{k_n} & \xrightarrow{\gamma_B} & B_{k_1+\cdots+k_n} \end{array}$$

However, it is clear that this simple generalization is not what we need. Indeed, one can immediately observe, for example, that the forgetful morphism $\varphi : \mathcal{MS} \rightarrow \mathcal{SO}$ that shrinks the morphological trees to their root vertices is not a morphism of graded algebras over the $\mathcal{M}$ operad, as it obviously does not preserve degrees, since all the leaves of each morphological tree are identified to the same leaf of the image in $\mathcal{SO}$.

Since we are interested not so much in this forgetful morphism but rather in the operation of *adding* morphology to syntactic trees, we formulate a generalization of morphisms of graded algebras over operads that will account for this transition from syntactic to morphosyntactic trees.

We extend the notion of morphisms of (graded) algebras over operads to a more flexible notion of *correspondences*. This replaces directly mapping the sets $\varphi_n : A_n \rightarrow B_n$, by maps involving auxiliary sets, which make it possible to consistently change degrees compatibly with the operad action.

Definition 3.14. A correspondence $\mathcal{C} = (C, \gamma_{A,C}) : A \rightarrow B$ between graded algebras $A = \cup_n A_n$ and $B = \cup_n B_n$ over an operad $\mathfrak{D}$ is a collection of sets $C = \cup_n C_n$ and maps

$$(3.15) \quad \gamma_{A,C} : A_n \times C_{k_1} \times \cdots \times C_{k_n} \rightarrow B_{k_1+\cdots+k_n}$$

for all $n, k_1, \dots, k_n \geq 1$, that satisfy the compatibility properties with the operad actions γ_A and γ_B given by commutative diagrams of the form

$$\begin{array}{ccccc} X & \xrightarrow{\gamma_A \times \text{id}} & Y & & \\ \downarrow \sigma & & \searrow \gamma_{A,C} & & \\ & & & Z & \\ & & \nearrow \gamma_B & & \\ U & \xrightarrow{\text{id} \times \gamma_{A,C}^n} & V & & \end{array}$$

with σ the permutation of the factors and

$$\begin{aligned} X &= \mathfrak{D}(n) \times A_{k_1} \times \cdots \times A_{k_n} \times C_{\ell_{1,1}} \times \cdots \times C_{\ell_{1,k_1}} \times \cdots \times C_{\ell_{n,1}} \times \cdots \times C_{\ell_{n,k_n}}, \\ Y &= A_{k_1+\cdots+k_n} \times C_{\ell_{1,1}} \times \cdots \times C_{\ell_{n,k_n}}, \\ U &= \mathfrak{D}(n) \times A_{k_1} \times C_{\ell_{1,1}} \times \cdots \times C_{\ell_{1,k_1}} \times \cdots \times A_{k_n} \times C_{\ell_{n,1}} \times \cdots \times C_{\ell_{n,k_n}}, \\ V &= \mathfrak{D}(n) \times B_{\ell_{1,1}+\cdots+\ell_{1,k_1}} \times \cdots \times B_{\ell_{n,1}+\cdots+\ell_{n,k_n}}, \\ Z &= B_{\ell_{1,1}+\cdots+\ell_{n,k_n}}. \end{aligned}$$

Here we do not require the operad $\mathfrak{D}$ to be a colored operad, although that may be the case when syntactic objects are filtered for consistent theta roles assignments and for well formed phases, as in [20] and [18]. However, we still do need a *colored version* of the correspondences of Definition 3.14.

Definition 3.15. Suppose given finite sets Ω_A , Ω_B and Ω_C . Assume given a correspondence between these two sets, in the form of a subset $\Gamma \subset \Omega_C \times \Omega$. Suppose A is a graded algebras over an ordinary (non-colored) operad $\mathfrak{D}$, as in Definition 3.12, with the property that, for all $n \geq 1$, the sets A_n decompose as $A_n = \sqcup_{a_1, \dots, a_n \in \Omega_A} A_{a_1, \dots, a_n}$, so that the maps (3.14) restrict to maps

$$(3.16) \quad \gamma_A : \mathfrak{D}(n) \times A_{a_1, 1, \dots, a_1, k_1} \times \dots \times A_{a_n, 1, \dots, a_n, k_n} \rightarrow A_{a_1, 1, \dots, a_n, k_n}.$$

We say that A is a graded colored algebra over $\mathfrak{D}$. A colored correspondence $\mathcal{C} = (C, \gamma_{A,C})$ of a graded colored algebras A and B over the operad $\mathfrak{D}$ is a collection of sets $C = \sqcup_n C_n$ with

$$C_n = \bigsqcup_{u \in \Omega_C} \bigsqcup_{b_1, \dots, b_n \in \Omega_B} C_{b_1, \dots, b_n}^u$$

with the property that the maps $\gamma_{A,C}$ of (3.15) restrict to maps

$$(3.17) \quad \gamma_{A,C} : A_{c_1, \dots, c_n} \times C_{b_1, 1, \dots, b_1, k_1}^{u_1} \times \dots \times C_{b_n, 1, \dots, b_n, k_n}^{u_n} \rightarrow B_{b_1, 1, \dots, b_n, k_n}$$

defined on the domains

$$(3.18) \quad \text{Dom}(\gamma_{A,C}) = \{(x, y_1, \dots, y_n) \mid x \in A_{c_1, \dots, c_n}, y_i \in C_{b_i, 1, \dots, b_i, k_i}^{u_i} \text{ with } (u_i, c_i) \in \Gamma\}.$$

The maps (3.17) satisfy compatibility with the maps γ_A and γ_B of graded colored algebras over $\mathfrak{D}$, as in (3.16), of the same form as the diagrams in Definition 3.14 with the colored decompositions of the sets A, B, C taken into account.

Theorem 3.16. The set $\widetilde{\mathcal{MO}}$ of extended morphological objects is a colored correspondence between the algebras over the Merge operad $\mathcal{M}$ given by the set $\mathcal{SO}$ of syntactic objects and the set $\mathcal{MS}$ of morphosyntactic trees, with color sets $\Omega_{\mathcal{SO}} = \mathcal{SO}_0$, $\Omega_{\mathcal{MS}} = \mathcal{P}(\mathcal{MO}_0)$, and $\Omega_{\widetilde{\mathcal{MO}}} = \mathcal{P}(\mathcal{MO}_0)$.

Proof. We just need to check that the maps $\gamma_{\mathcal{SO}}$ that give the Merge operad action on syntactic objects and $\gamma_{\mathcal{MS}}$, the operad action on morphosyntactic trees, are compatible with the maps $\gamma_{\mathcal{SO}, \mathcal{MO}}$ of (3.10) through commutative diagrams as in Definition 3.14, for the colored version as in Definition 3.15. This is the case since the colors $\Omega_{\mathcal{MS}} = \mathcal{P}(\mathcal{MO}_0)$ and $\Omega_{\widetilde{\mathcal{MO}}} = \mathcal{P}(\mathcal{MO}_0)$ are the feature bundles (or single features) at the leaves of both extended morphological objects and morphosyntactic trees and the colors $\Omega_{\mathcal{SO}} = \mathcal{SO}_0$ are the lexical items at the leaves of the syntactic objects and the operad insertions of roots of extended morphological objects at leaves of syntactic trees that forms morphosyntactic trees is constrained by the Syntax-Morphology feature correspondence of Definition 2.4 that gives the domains as in (3.18). The compatibility of all the maps in the diagrams then follows. $\square$

4. STRUCTURE BUILDING IN MORPHOSYNTAX

We now combine the structures we have discussed so far involving morphological objects and their interfacing with syntactic objects. In particular, we combine the existence of the coproduct Δ^ρ on the span of morphological workspaces with the operadic insertion maps $\gamma_{\mathcal{SO}, \mathcal{MO}}$ of (3.10) relating morphological objects, syntactic objects and morphosyntactic trees.

In syntax, Merge, in the form (1.2) or in the assembled form (1.3), is the fundamental structure building operation. In the interface between syntax and morphology, structure building in syntax has been completed, hence the syntactic objects $T \in \mathcal{SO}$ are available as material for the structure building of morphosyntax. Each syntactic object $T \in \mathcal{SO}$

contributes a structure building operation $\mathcal{K}_T$ that takes material from the morphological data (the morphological workspaces) and assembles corresponding morphosyntactic trees, using the operadic insertions $\gamma_{\mathcal{SO}, \mathcal{MO}}$ described above. The key property of the operation $\mathcal{K}_T$ is that it combines the insertion operations $\gamma_{\mathcal{SO}, \mathcal{MO}}$ with the product $\sqcup$ and coproduct Δ^ρ structure of the Hopf algebra of morphological workspaces.

Definition 4.1. Let $\mathcal{W}_{\mathcal{MS}} = \mathfrak{F}(\mathcal{MS})$ denote the set of morphosyntactic workspaces, the set of forests whose components are morphosyntactic trees, and let $\mathcal{V}(\mathcal{W}_{\mathcal{MS}})$ be the vector space spanned by these forests. Consider the following linear maps on $\mathcal{K}_T : \mathcal{V}(\mathcal{W}_{\mathcal{M}}) \rightarrow \mathcal{V}(\mathcal{W}_{\mathcal{MS}})$, for $T \in \mathcal{SO}$ a syntactic object with data $\alpha_\ell \in \mathcal{SO}_0$ at the leaves $\ell \in L = L(T)$, defined as

$$(4.1) \quad \mathcal{K}_T := \sqcup \circ (\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots) \otimes \text{id}) \circ \delta_{\underline{B}, \underline{\alpha}_L} \circ \Delta^\rho,$$

where $\delta_{\underline{B}, \underline{\alpha}_L}^B$, for $B = (B_\ell)_{\ell \in L}$ and $\underline{\alpha}_L = (\alpha_\ell)_{\ell \in L}$, is the linear operator defined on the basis elements as follows and then extended by linearity:

$$\delta_{\underline{B}, \underline{\alpha}_L}(F_{\underline{v}} \otimes F/\rho F_{\underline{v}}) = \begin{cases} F_{\underline{v}} \otimes F/\rho F_{\underline{v}} & F_{\underline{v}} = \sqcup_{\ell \in L} T_\ell \text{ with } B_{v_\ell} = B_\ell \text{ and } (B_\ell, \alpha_\ell) \in \Gamma_{SM} \\ 0 & \text{otherwise.} \end{cases}$$

where $B_{v_\ell} \in \mathcal{P}(\mathcal{MO}_0)$ is the bundle of morphological features assigned to the root v_ℓ of the extended morphological tree T_ℓ that will be inserted at the leaf $\ell \in L = L(T)$ of the syntactic tree T .

The operations $\mathcal{K}_T$ defined as in (4.1) resemble the form of the Merge operation in (1.3), hence we are using a similar notation.

We can also formulate an analog of the Merge operations $\mathfrak{M}_{S, S'}$ of (1.2), for which we will also use a similar notation $\mathfrak{M}_{S_1, \dots, S_n}^T$ (see (4.2) below). These operations isolate individual terms of $\mathcal{K}_T$, as in (4.3), which is an analog of the sum in (1.3). While formally these operations look similar to their syntactic counterparts, there are important differences:

- Unlike the syntactic Merge of (1.2) and (1.3), the operations $\mathcal{K}_T$ and $\mathfrak{M}_{S_1, \dots, S_n}^T$ use already formed syntactic and morphological objects to assemble morphosyntactic objects, hence they do not map to the same space, hence they cannot define a dynamical system by iteration (unlike the Hopf algebra Markov chain of Merge, that gives the syntactic derivations).
- The operations $\mathcal{K}_T$ and $\mathfrak{M}_{S_1, \dots, S_n}^T$ are post-syntactic, in the sense that they rely on the products of syntactic Merge and model the interface of syntax with morphology.

For $T \in \mathcal{SO}_n$ and $S_1, \dots, S_n \in \widetilde{\mathcal{MO}}$ such that, for all $\ell \in L(T)$ the pair (B_ℓ, α_ℓ) is in Γ_{SM} for B_ℓ the morphological feature bundle at the root of S_ℓ and $\alpha_\ell \in \mathcal{SO}_0$ at the corresponding leaf of T , we set

$$(4.2) \quad \mathfrak{M}_{S_1, \dots, S_n}^T = \sqcup \circ (\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots) \otimes \text{id}) \circ \delta_{S_1, \dots, S_n} \circ \Delta^\rho,$$

where the linear operator $\delta_{S_1, \dots, S_n}$ is defined on the basis elements as follows and extended by linearity:

$$\delta_{S_1, \dots, S_n}(F_{\underline{v}} \otimes F/\rho F_{\underline{v}}) = \begin{cases} F_{\underline{v}} \otimes F/\rho F_{\underline{v}} & F_{\underline{v}} = \sqcup_{\ell} S_\ell \\ 0 & \text{otherwise.} \end{cases}$$

These satisfy

$$(4.3) \quad \mathcal{K}_T = \sum_{S_1, \dots, S_n} \mathfrak{M}_{S_1, \dots, S_n}^T$$

where the sum is taken over all $S_1, \dots, S_\ell$ satisfying the conditions $(B_\ell, \alpha_\ell) \in \Gamma_{SM}$. While the right-hand side is formally an infinite sum, it is always a finite sum when applied to a given morphological workspace. These operations allow us to extract terms from the morphological workspaces via Δ^ρ and insert them at the leaves of syntactic trees, resulting in morphosyntactic trees.

Definition 4.2. *We denote by*

$$(4.4) \quad \mathcal{L}_{\mathcal{SO}, \mathcal{MO}} = \{\mathfrak{M}_{S_1, \dots, S_n}^T \mid T \in \mathcal{SO}, S_1, \dots, S_n \in \widetilde{\mathcal{MO}}\}$$

the set of all the linear operators of the form (4.2) that interface syntactic objects with morphological trees building resulting morphosyntactic structures.

We will see in §5 certain well known operations of Distributed Morphology can be seen as transformations acting on this set $\mathcal{L}_{\mathcal{SO}, \mathcal{MO}}$.

5. THE OPERATIONS OF DISTRIBUTED MORPHOLOGY

The four DM operations of fission, fusion, impoverishment, and obliteration operate on morphosyntactic trees as post-syntactic operations which manipulate the morphosyntactic tree structures. We now approach the mechanics of the four DM operations. We will present them in two different but equivalent perspectives. First we reformulate in our setting the usual way in which these operations are described in the DM literature, by presenting them as transformations of morphosyntactic trees. It should be pointed out that they are often seen just as transformations of morphological trees (or bundles of features), but in fact the way the morphological structures are inserted at the leaves of syntactic trees matters in defining these operations, as will be clear in the following, so they should be regarded as acting on fully formed morphosyntactic trees. There is, however, another equivalent viewpoint that we will present, that identifies these operations of DM as transformations acting on the set of all the structure-building operations $\{\mathfrak{M}_{S_1, \dots, S_n}^T\}$ of morphosyntax. In this perspective, the DM operations are not so much altering morphological or morphosyntactic trees, but rather altering the recipes for assembling morphosyntactic trees.

As we will see more in detail below, fusion pushes the morphological part of morphosyntax upward into the syntactic part (by changing a syntactic vertex of the morphosyntactic tree into a morphological vertex), while fission does the opposite operation, pushing syntax downward into the morphology part, transforming a morphological vertex into a syntactic vertex. While these two operations appear in this sense symmetric there is a key difference in their formalization: fusion only uses the magma operation that is common to both morphology and syntax, while fission is a truly morphological operation that also involves set-theoretic operations on bundles of morphological features and could not exist within syntax. This difference is not surprising: since the fusion operation moves toward syntax, it should rely only on operations that are available within syntax, while fission that moves toward morphology relies also on purely morphological data. The remaining two operations, obliteration and impoverishment, rely on the coproduct Δ^ρ and (in the case of impoverishment) on the fission operation.

5.1. Fusion. Fusion refers to two different morphological feature bundles in two different adjacent syntactic leaves (two leaves of a cherry subtree) being merged into one feature bundle. This can apply, for instance, to the case of two different adjacent heads that are merged together as derived from head-to-head movement (for a discussion of this type of

head-to-head movement in the context of the larger mathematical formulation see [19]). This fusion mechanism is typically thought to also combine the two leaves together, so that the resulting feature bundle is assigned to the single remaining leaf vertex. That is, the syntactic tree itself is also modified, with a cherry tree contracted to its root vertex, which becomes the new leaf.

An explicit example of fusion can be seen in negation in Swahili [23].

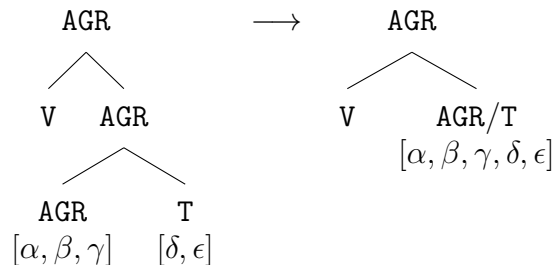
Example 5.1. In the case of 1PL in Swahili, “We will love Swahili” and its negation “We will not love Swahili” are stated in (a) and (b), respectively:

- a. tu- ta- pend-a kiswahili
we- will- love Swahili
‘We will love Swahili’.
- b. ha- tu- ta- pend-a kiswahili
neg- we- will- love Swahili
‘We will not love Swahili’.

However, the same is not the case for 1SG subjects. In particular, 1SG “I will love Swahili” is expressed with the 1SG prefix in place of the 1PL in (a) above, negation cannot be expressed by an identical replacement of the 1SG into 1PL in (b) above. That expression of “I will not love Swahili” is ungrammatical—instead, the negation and subject 1SG must be expressed together in the single morpheme *si-*. These three cases in 1SG are demonstrated as (a-c) below:

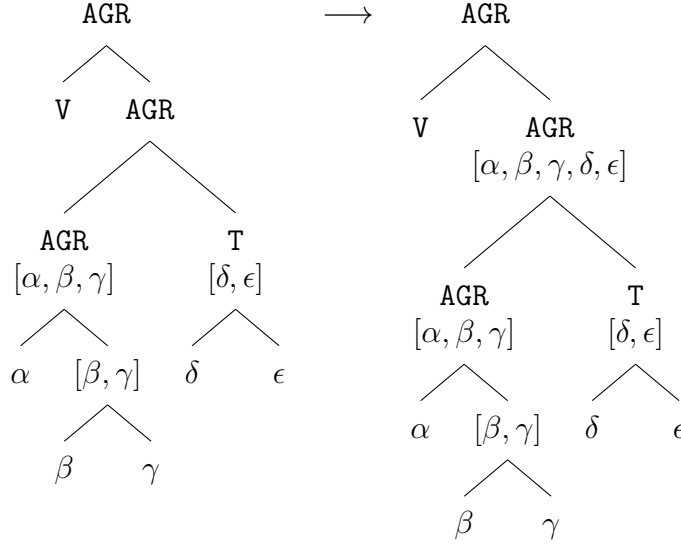
- a. ni- ta- pend-a kiswahili
I- will- love Swahili
‘I will love Swahili’.
- b. *ha- ni- ta- pend-a kiswahili
neg- I- will- love Swahili
Intended: ‘I will not love Swahili’.
- c. si- ta- pend-a kiswahili
not.I- will- love Swahili
‘I will not love Swahili’.

The process of fusion is depicted in [4] in the following way, where the structure on the left has the two leaves AGR and T merged into one:



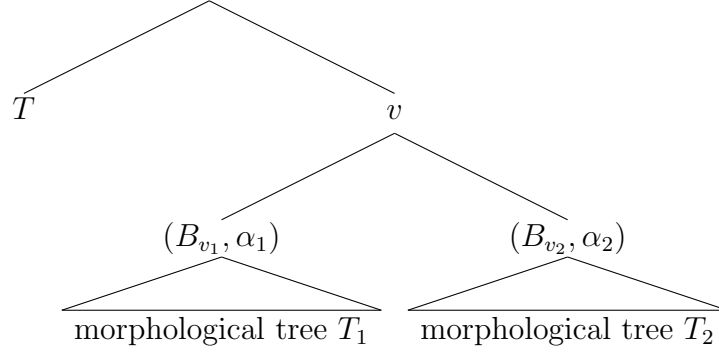
The label AGR/T represents the fact that, with this formulation, it is unclear how the new leaf that has fused the two original leaves with different heads should be labeled. We will see below that this labeling issue can in fact be easily resolved.

We show that fusion can be formulated in our setting in a way that does *not* require modifying the underlying tree structure. We depict Fusion then as the following, where the feature bundles are being represented as their hierarchical tree structures.

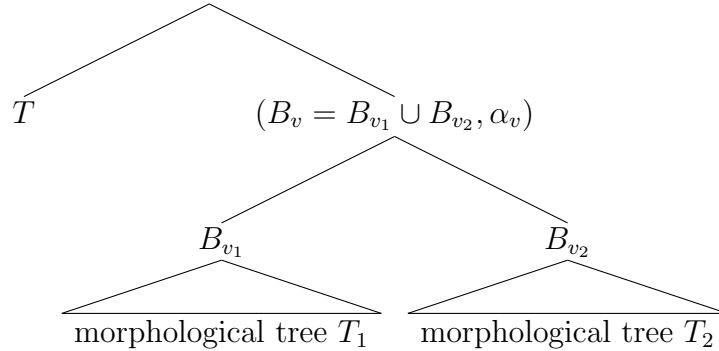


The idea here is that we reassemble the morphosyntactic tree by inserting an enlarged morphological tree at a leaf of a reduced syntactic tree, provided a matching condition holds, according to the matching rule Γ_{SM} of Definition 2.4.

This means that, more generally, we represent fusion as the operation that transforms a morphosyntactic tree of the form



into a morphosyntactic tree of the form



where the vertex v , that is part of a syntactic object in the first tree, becomes part of a morphological object in the second tree, with the associated feature bundle $B_v = B_{v_1} \cup B_{v_2}$. Note that the vertices v_1 and v_2 , in the first tree, are the vertices where the insertion operation $\gamma_{\mathcal{SO}, \mathcal{MO}}$ has attached morphological data to the leaves of a syntactic tree, hence they are also decorated with the corresponding data $\alpha_1, \alpha_2 \in \mathcal{SO}_0$ that were at the leaves of the syntactic

tree, with the condition that $(B_{v_i}, \alpha_i) \in \Gamma_{SM}$ so that the insertion $\gamma_{\mathcal{SO}, \mathcal{MO}}$ can take place. In the second tree, the vertices v_1, v_2 become interior vertices of a morphological tree, hence they still carry the B_{v_i} labels but they no longer carry the syntactic α_i labels. On the other hand, the vertex v has now become a leaf for the syntactic tree and the place where the insertion $\gamma_{\mathcal{SO}, \mathcal{MO}}$ takes place, so it needs to carry also a datum $\alpha_v \in \mathcal{SO}_0$. The way to obtain this label is via the head function $h_{T'}$ of the syntactic tree T' and the labeling algorithm for syntactic objects. Note that our choice to label the resulting syntactic vertex with $\alpha_v = \alpha_{h_{T'}(v)}$ seems to completely remove the information coming from the other α_i that does not project. This is not the case, however, since we must have the condition $(B_{v_1} \cup B_{v_2}, \alpha_v) \in \Gamma_{SM}$ in order to still perform the matching at v of syntactic and morphological data. In the tree we start with, we have $(B_{h_{T'}(v)}, \alpha_{h_{T'}(v)}) \in \Gamma_{SM}$ and also $(B_i, \alpha_i) \in \Gamma_{SM}$ for the other leaf, but these two separate conditions do not a priori imply $(B_{v_1} \cup B_{v_2}, \alpha_v) \in \Gamma_{SM}$. The fact that this holds (which is a necessary condition for fusion to take place) can be interpreted as a relation between α_1, α_2 that makes it possible to match both B_{v_1} and B_{v_2} to α_v .

Definition 5.2. *The fusion operation is a linear operator $\mathcal{F} : \mathcal{V}(\mathcal{W}_{MS}) \rightarrow \mathcal{V}(\mathcal{W}_{MS})$ on the space of morphosyntactic trees, extended by linearity and defined on basis elements as $\mathcal{F}(F) = \sqcup_i \mathcal{F}(T_i)$ for $F = \sqcup_i T_i \in \mathcal{W}_{MS}$, where for $T \in \mathcal{MS}$ we define $\mathcal{F}(T)$ in the following way. The morphosyntactic tree is of the form*

$$T = \gamma_{\mathcal{SO}, \mathcal{MO}}(T', (S_\ell)_{\ell \in L(T')}), \quad \text{for } T' \in \mathcal{SO} \quad \text{and } S_\ell \in \widetilde{\mathcal{MO}}.$$

We define

$$(5.1) \quad \mathbf{C}(T) := \{T_v \in \text{Acc}(T) \mid T_v = \begin{array}{c} v \\ \swarrow \quad \searrow \\ (B_{v_1}, \alpha_1) \quad (B_{v_2}, \alpha_2) \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \text{tree } S_1 \quad \text{tree } S_2 \end{array} \}$$

namely the set of accessible terms of the morphosyntactic trees that are cherry trees $\mathfrak{M}(\alpha_1, \alpha_2)$ of a syntactic tree with morphological insertions at both leaves, with $(B_{v_i}, \alpha_i) \in \Gamma_{SM}$. We then set

$$(5.2) \quad \mathcal{F}(T) = \sum_{T_v \in \mathbf{C}(T)} \gamma_{\mathcal{SO}, \mathcal{MO}}(T'/^c \mathfrak{M}(\alpha_1, \alpha_2), S_{12}, (S_\ell)_{\ell \in L(T') \setminus \{\ell_1, \ell_2\}}),$$

where ℓ_1, ℓ_2 are the leaves of T' marked by $\alpha_1, \alpha_2 \in \mathcal{SO}_0$ and $S_{12} \in \widetilde{\mathcal{MO}}$ is given by

$$(5.3) \quad S_{12} = \begin{array}{c} (B_v = B_{v_1} \cup B_{v_2}, \alpha_v) \\ \swarrow \quad \searrow \\ B_{v_1} \quad B_{v_2} \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \text{tree } S_1 \quad \text{tree } S_2 \end{array}$$

where $\alpha_v = \alpha_{h_{T'}(v)} \in \{\alpha_1, \alpha_2\}$.

Note that, here and in the following, we write for simplicity S_1 and S_2 as the morphological objects fusing to S_{12} , though they can just be any pair $S_\ell, S_{\ell'}$ attached to the same syntactic vertex v in the morphosyntactic tree.

The expression $\mathcal{F}(T)$ of (5.2) has the effect of removing the accessible term $\mathfrak{M}(\alpha_1, \alpha_2)$ of the syntactic object T' , leaving a leaf in place of the root vertex v of $\mathfrak{M}(\alpha_1, \alpha_2)$ (which means using the quotient $T'/^c \mathfrak{M}(\alpha_1, \alpha_2)$ that shrinks $\mathfrak{M}(\alpha_1, \alpha_2)$ to its root vertex, and then, instead of

$T = \gamma_{\mathcal{SO}, \mathcal{MO}}(T', S_1, S_2, \dots, S_n)$, one computes $\gamma_{\mathcal{SO}, \mathcal{MO}}(T' / {}^c\mathfrak{M}(\alpha_1, \alpha_2), S_{12}, S_3, \dots, S_n)$. This performs the change that we described above, and the sum over $\mathbf{C}(T)$ means that one considers (as a sum) all the possibilities where this operation can be performed in the given T (and similarly for a whole workspace $F \in \mathcal{W}_{\mathcal{MS}}$).

Just as for Merge in syntax we can decompose $\mathcal{K} = \sum_{S, S'} \mathfrak{M}_{S, S'}$ and for the operations that assemble morphosyntactic trees we can decompose $\mathcal{K}_T = \sum_{S_1, \dots, S_n} \mathfrak{M}_{S_1, \dots, S_n}^T$, as discussed above, we can also similarly decompose

$$(5.4) \quad \mathcal{F} = \sum_{S_1, S_2} \mathcal{F}_{\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)},$$

where the transformations $\mathcal{F}_{\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)}$ target a specific morphosyntactic subtree given by the insertion $\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)$ of two morphological trees S_1, S_2 with root feature bundles B_{v_1} and B_{v_2} into the two leaves of a syntactic object $\mathfrak{M}(\alpha_1, \alpha_2)$ with $(B_{v_i}, \alpha_i) \in \Gamma_{SM}$. This map acts on a workspace $F \in \mathcal{W}_{\mathcal{MS}}$ by searching for a copy of the object $\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)$ among the accessible terms of the workspace, and replacing it with the object $\gamma_{\mathcal{SO}, \mathcal{MO}}((B_v, \alpha_v), S_{12})$ where (B_v, α_v) is the root vertex v of $\mathfrak{M}(\alpha_1, \alpha_2)$ with this labeling. This can also be reformulated through a characterization of the fusion transformation in the following way.

Proposition 5.3. *The fusion operation $\mathcal{F}$ of (5.2) and (5.4) can be characterized as the unique map that makes the following diagram commute:*

$$\begin{array}{ccc} \mathcal{V}(\mathcal{W}_{\mathcal{M}}) & \xrightarrow{\mathfrak{M}_{S_1, S_2, \dots, S_n}^T} & \mathcal{V}(\mathcal{W}_{\mathcal{MS}}) \\ \mathfrak{M}_{S_1, S_2}^{\text{morph}} \downarrow & & \downarrow \mathcal{F}_{\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)} \\ \mathcal{V}(\mathcal{W}_{\mathcal{M}}) & \xrightarrow{\mathfrak{M}_{S_{12}, \dots, S_n}^{T / {}^c\mathfrak{M}(\alpha_1, \alpha_2)}} & \mathcal{V}(\mathcal{W}_{\mathcal{MS}}) \end{array}$$

We can interpret Proposition 5.3 as providing a different but equivalent viewpoint on the fusion operation $\mathcal{F}$. The description we gave in terms of the map $\mathcal{F}_{\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)} : \mathcal{V}(\mathcal{W}_{\mathcal{MS}}) \rightarrow \mathcal{V}(\mathcal{W}_{\mathcal{MS}})$ is a transformation of morphosyntactic trees. The corresponding map $\mathfrak{M}_{S_1, S_2}^{\text{morph}}$ describes what happens if one sees fusion as a transformation of morphological trees (as it is often described in DM). The two are related via the morphosyntax-assembly operations $\mathfrak{M}_{S_1, S_2, \dots, S_n}^T$ and $\mathfrak{M}_{S_{12}, \dots, S_n}^{T / {}^c\mathfrak{M}(\alpha_1, \alpha_2)}$. This is one way of reading the commutative diagram of Proposition 5.3 (in other words, to read it “horizontally”: the horizontal maps relate “fusion as an operation in morphology” and “fusion as an operation in morphosyntax”). However, there is another way of reading the same diagram, namely reading it “vertically”. When seen in this way, the two vertical arrows are a transformation between the two horizontal arrows, or in other words a map that changes a morphosyntax-assembly operation $\mathfrak{M}_{S_1, S_2, \dots, S_n}^T$ into another one, $\mathfrak{M}_{S_{12}, \dots, S_n}^{T / {}^c\mathfrak{M}(\alpha_1, \alpha_2)}$. This description shows that we can also think of fusion as acting on the set of operations $\{\mathfrak{M}_{S_1, \dots, S_n}^T\}$. As we will see below, this is the case also for the other operations of DM.

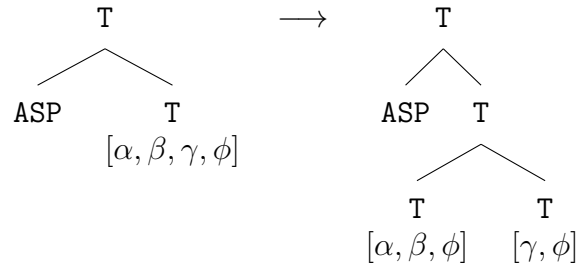
5.2. Fission. Fission is the process of splitting one feature bundle into two. We can motivate the existence of fission by the fact that there is reason to believe that two features exist in one single feature bundle within the syntax, but are realized as separate phonological exponents. Consider the following data from Arabic, as given by [15] originally from [28, p.56].

Example 5.4. *Ṣanṣānī* Arabic has discontinuous agreement of person and number in the context of prefix conjugation. With the standard assumption in syntax that the subject's morphological features are all housed in a single syntactic head, which includes both person and number, only first person consistently has these features realized as a single affix (the prefix). Second and third person exemplify discontinuous agreement: the person and number are realized as both prefixes and suffixes on the verb. For example, the verb *gmbr*, 'sit', is conjugated as follows, where the person agreement affixes are indicated in bold:

	SG	PL
1	ʔa -gambir	ni -gambir
2M	ti -gambir	ti -gambir- ū
2F	ti -gambir- ī	ti -gambir- ayn
3M	yi -gambir	yi -gambir- ū
2F	ti -gambir	yi -gambir- ayn

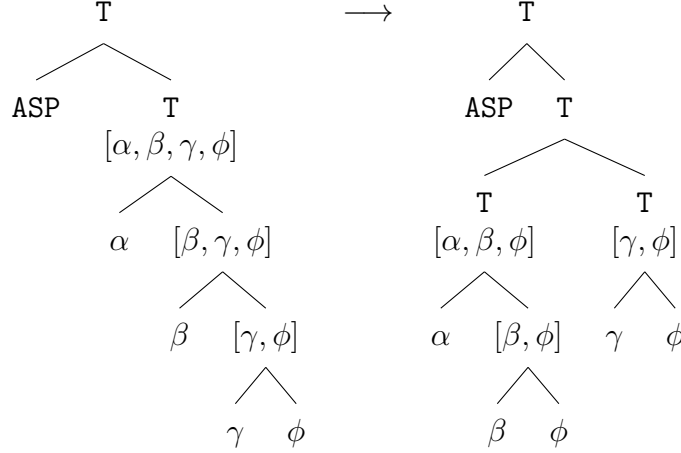
The formulation of [15] presents the idea that the two subsets of features to be fissioned from each other are each partitioned into a separate feature bundle, but the remaining features, indicated by ϕ , are copied into *both* fissioned bundles. There is reason to think that the non-fissioned features exist in both places, because they sometimes are pronounced multiple times, in each of the two vocabulary items corresponding to the two feature bundles resulting from fission. Hence ϕ , the remaining features irrelevant for fission, should appear in both feature bundles.

Suppose that α and β are to be fissioned from γ . This can be depicted as follows, where ϕ represents one or more additional features:



If we depict this with our tree interpretation of feature bundles, this becomes the following example.

Example 5.5. A fission operation (with fissioned feature ϕ) is illustrated by the example:



As observed in Remark 2.2, we can realize a bundle B_v of features at the root of a morphological tree $T \in \mathcal{MO}$ through alternative tree structures where some of the features $\phi \in B_v$ are repeated in lower vertices of the tree not on the same root-to-leaf path. The fission operations can be seen as transformations that alter the tree decomposition of a certain bundle of morphological features, while at the same time replacing a morphological vertex with a syntactic one, hence lowering the boundary between syntax and morphology, unlike fusion which raises it. We will discuss this more in §6.

Again we define the fission operations as linear transformations on the space $\mathcal{V}(\mathcal{W}_{\mathcal{MS}})$ of morphosyntactic trees, by defining the action on a single morphosyntactic tree, extending it multiplicatively (in the $\sqcup$ product) on forests and additively on linear combinations, as we did for the case of fusion. Also, as in the case of fusion, we define the operator that performs all the possible fission operations on a given tree (or forest) and presents the results as a sum of each individual fission (which is the analog of the $\mathcal{F}(T)$ fusion of (5.2)), and then we decompose it into individual fission operations (as in (5.4) for the case of fusion).

Consider a morphological tree $S \in \widetilde{\mathcal{MO}}$ and let B be the bundle of morphological features at the root vertex v of S . For any subset of this feature bundle, $A \in \mathcal{P}(B)$, consider the set $B \setminus A$ (the subset of B not including any elements of the set A) and the set of all partitions

$$(5.5) \quad \mathfrak{P}_{A, \alpha_1, \alpha_2}(B) := \{(B_1, B_2) \mid B \setminus A = B_1 \sqcup B_2 \text{ and } (B_i \cup A, \alpha_i) \in \Gamma_{SM}\}.$$

Definition 5.6. Consider a given $T \in \mathcal{MS}$ with $T = \gamma_{\mathcal{SO}, \mathcal{MO}}(T', S_1, \dots, S_n)$ for $T' \in \mathcal{SO}$ and $S_\ell \in \widetilde{\mathcal{MO}}$, for $\ell \in L(T')$, with $(B_{v_\ell}, \alpha_\ell) \in \Gamma_{SM}$ for v_ℓ the root vertex of S_ℓ that $\gamma_{\mathcal{SO}, \mathcal{MO}}$ inserts at the ℓ leaf of T' . We define the fission operation as

$$(5.6) \quad \Phi(T) = \sum_{\ell, \alpha, A, (B_1, B_2)} \Phi_{A, (B_1, B_2), \alpha}(T) \quad \text{with} \quad \begin{array}{l} \ell \in L(T'), \quad \alpha \in \mathcal{SO}_0, \\ A \in \mathcal{P}(B_{v_\ell}), \quad (B_1, B_2) \in \mathfrak{P}_{A, \alpha_\ell, \alpha}(B_{v_\ell}), \end{array}$$

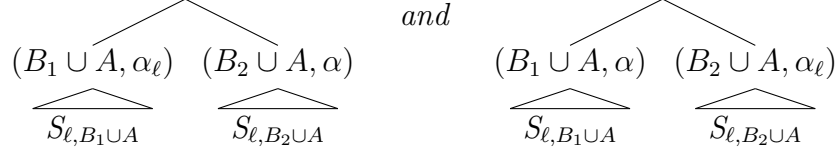
$$(5.7) \quad \Phi_{A, (B_1, B_2), \alpha}(T) = \gamma_{\mathcal{SO}, \mathcal{MO}}(T' \circ_{\mathcal{SO}_\ell} \mathfrak{M}(\alpha_\ell, \alpha), S_1, \dots, S_{\ell, B_1 \cup A}, S_{\ell, B_2 \cup A}, \dots, S_n),$$

where the operation $\circ_{\mathcal{SO}_\ell}$ is the single leaf insertion as in (3.8), and the two morphological trees $S_{\ell, B_i \cup A}$ are obtained from S_ℓ by performing the following operations:

- replacing B_{v_ℓ} at the root with $B_{v_i} = B_i \cup A$,
- for each vertex w of S_ℓ below the root, take $B_w \cap (B_i \cup A)$
- in the resulting tree consider the subforest $F_{\underline{w}}$ whose components S_{w_i} are the accessible terms of S_ℓ with root vertex w_i (and hence all other vertices as well) satisfying $B_{w_i} \cap (B_i \cup A) = \emptyset$

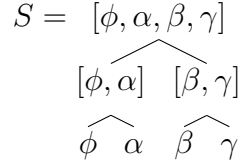
- take $S_{\ell, B_i \cup A} = S_{\ell} / {}^{\rho} F_{\underline{w}}$ with the vertices w labelled by $B_w \cap (B_i \cup A)$.

Since $\mathfrak{M}(\alpha_{\ell}, \alpha)$ is non-planar, in the insertion of $S_{\ell, B_1 \cup A}, S_{\ell, B_2 \cup A}$ at the leaves of $\mathfrak{M}(\alpha_{\ell}, \alpha)$ both possibilities (differing in the assignment of the head function) are included in the sum:



The procedure described here is the general form of what we have seen in Example 5.5.

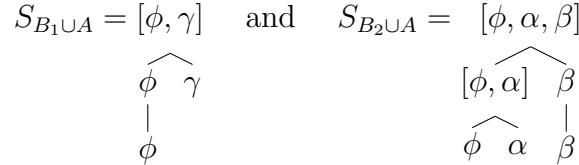
Example 5.7. For example, consider the case of a morphological tree of the form



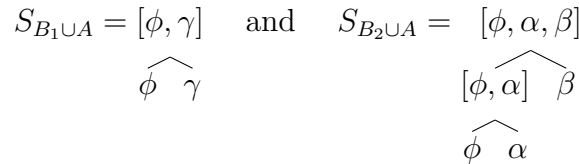
and take $B_1 \cup A = [\phi, \gamma]$ and $B_2 \cup A = [\phi, \alpha, \beta]$ with $A = \phi$. The procedure described above for the construction of the trees $S_{B_i \cup A}$ starts with producing the list of feature bundles $B_w \cap (B_i \cup A)$,

	$B_w \cap (B_1 \cup A)$	$B_w \cap (B_2 \cup A)$
level 2	$[\phi, \alpha] \cap [\phi, \gamma] = \phi$	$[\phi, \alpha] \cap [\phi, \alpha, \beta] = [\phi, \alpha]$
level 2	$[\beta, \gamma] \cap [\phi, \gamma] = \gamma$	$[\beta, \gamma] \cap [\phi, \alpha, \beta] = \beta$
level 3	$\phi \cap [\phi, \gamma] = \phi \quad \alpha \cap [\phi, \gamma] = \emptyset$	$\phi \cap [\phi, \alpha, \beta] = \phi \quad \alpha \cap [\phi, \alpha, \beta] = \alpha$
level 3	$\beta \cap [\phi, \gamma] = \emptyset \quad \gamma \cap [\phi, \gamma] = \emptyset$	$\beta \cap [\phi, \alpha, \beta] = \beta \quad \gamma \cap [\phi, \alpha, \beta] = \emptyset$

which gives the trees



The two non-branching vertices can be eliminated as they do not add any features, resulting in the trees



In order to characterize fission with a commutative diagram akin to the commutative diagram we gave for fusion, we introduce the *root-cut operator*.

Definition 5.8. The root-cut operator from trees to forests is defined by

$$(5.8) \quad \mathfrak{C}(T) = T_1 \sqcup \cdots \sqcup T_n, \quad \text{for } T = \mathfrak{B}(T_1 \sqcup \cdots \sqcup T_n).$$

Namely, it performs the opposite operation of the grafting $\mathfrak{B}$: instead of appending all the component trees of a forest to a common root, forming a single tree, it cuts all the edges below the root of a tree, resulting in a forest. In particular, if $\mathfrak{T}_X$ is the set of non-planar

binary rooted trees with leaves labelled by elements of a set X and $\mathfrak{F}_X$ is the set of forests with components in $\mathfrak{T}_X$,

$$(5.9) \quad \mathfrak{C}(T) = T_1 \sqcup T_2, \quad \text{for } T = \mathfrak{M}(T_1, T_2) = \mathfrak{B}(T_1 \sqcup T_2).$$

In particular, for the case of morphological trees, we write

$$\mathfrak{C}_S^{\text{morph}} : \mathcal{V}(\mathcal{W}_{\mathcal{M}}) \rightarrow \mathcal{V}(\mathcal{W}_{\mathcal{M}})$$

for the map of morphological workspaces that acts as the root-cut operation on a component S of the workspace and as the identity on the other components,

$$(5.10) \quad \mathfrak{C}_S^{\text{morph}}(S \sqcup F) = \mathfrak{C}(S) \sqcup F = S_1 \sqcup S_2 \sqcup F, \quad \text{for } S = \mathfrak{M}^{\text{morph}}(S_1, S_2).$$

We consider the following family of maps of a similar type:

$$(5.11) \quad \mathfrak{C}_{A, (B_1, B_2)}^S(S \sqcup F) = S_{B_1 \cup A} \sqcup S_{B_2 \cup A} \sqcup F,$$

for $S_{B_1 \cup A}, S_{B_2 \cup A}$ constructed as above.

We can then give a characterization of fission similar to the one we gave of fusion in Proposition 5.3.

Proposition 5.9. *The fission operation Φ of (5.6) and (5.7) can be characterized as the unique map that makes the following diagram commute:*

$$\begin{array}{ccc} \mathcal{V}(\mathcal{W}_{\mathcal{M}}) & \xrightarrow{\mathfrak{M}_{S_1, \dots, S_\ell, \dots, S_n}^T} & \mathcal{V}(\mathcal{W}_{\mathcal{M}S}) \\ \mathfrak{C}_{A, (B_1, B_2)}^S \downarrow & \mathfrak{M}_{S_1, \dots, S_\ell, B_1 \cup A, S_\ell, B_2 \cup A, \dots, S_n}^{T \circ \ell \mathfrak{M}(\alpha_\ell, \alpha)} & \downarrow \Phi_{A, (B_1, B_2), \alpha} \\ \mathcal{V}(\mathcal{W}_{\mathcal{M}}) & \xrightarrow{\quad} & \mathcal{V}(\mathcal{W}_{\mathcal{M}S}) \end{array}$$

In the case where $A = \emptyset$ and B_1, B_2 are the labels of the vertices v_1, v_2 of the subtrees S_{v_1}, S_{v_2} with $S_\ell = \mathfrak{M}^{\text{morph}}(S_{v_1}, S_{v_2})$, the arrow $\mathfrak{C}_{A, (B_1, B_2)}^S$ is just the same as the root cut $\mathfrak{C}_S^{\text{morph}}$ of (5.8), in the form (5.10).

Again Proposition 5.9 provides us with two equivalent interpretations of fission: one as discussed above, as transformations of morphosyntactic and morphological trees, and the other as transformations acting on the set of operations $\{\mathfrak{M}_{S_1, \dots, S_n}^T\}$ of morphosyntax formation.

5.3. Obliteration. Obliteration is the complete removal of a feature bundle.

Example 5.10. As discussed in [1], the Ondarru dialect of Basque displays obliteration. Specifically, a first-person clitic is deleted when followed by a first- or second-person ergative clitic. This is described in [16] as the following rule:

- Within an auxiliary M-word with two clitics, c_1 and c_2 , delete c_1 when c_1 's feature bundle has [+participant, +author] and c_2 's feature bundle has [ergative, +participant].

Remark 5.11. The high frequency of these DM operations as being triggered by particular (i.e., language-specific) combinations of features suggests that these may correspond to filtering at the Externalization phase by language-dependent coloring rules. For example, one Externalization coloring rule pertaining to Ondarru's rule given in 5.10 would be to filter out any tree structures where the two colors $\mathbf{c}_1$ and $\mathbf{c}_2$ corresponding to c_1 and c_2 are adjacent/part of the same auxiliary M-word and c_1 and c_2 's feature bundles contained [+participant, +author] features and [ergative, +participant] features, respectively.

Obliteration hence corresponds to removing an entire morphological feature tree. This is motivated by any morphological case where an entire feature bundle is realized phonologically in some cases, but is not realized at all in others. In this case one can posit that the feature bundle was deleted and hence inaccessible to the usual vocabulary insertion rules that would normally realize that feature bundle, in other (morphosyntactic) circumstances.

In order to model obliteration within our formalism, we need to consider operations of the form $\mathfrak{M}_{S_1, \dots, S_n}^T$, where one or more of the $\{S_\ell\}_{\ell \in L(T)}$ are *empty*. This corresponds to cases where a syntactic leaf plays a role in the syntactic structure but does not carry an associated bundle of morphological features.

Remark 5.12. Allowing for some empty morphological insertions $S = \mathbf{1}$ requires weakening the assumptions that we made regarding the matching rules, specified by the correspondence Γ_{SM} of Definition 2.4, by dropping the surjectivity assumption that $\pi_2|_{\Gamma_{SM}} : \Gamma_{SM} \twoheadrightarrow \mathcal{SO}_0$. Indeed, if the map $\pi_2|_{\Gamma_{SM}}$ is not surjective, the set $\mathcal{SO}_0 \setminus \pi_2(\Gamma_{SM})$ represents the set of lexical items and syntactic features at which obliteration can happen.

Let us denote by $\mathbf{1}$ the unit of the magma of morphological objects, namely the formal empty tree. This is also the unit of the multiplication $\sqcup$ of the Hopf algebra of morphological workspaces. We can then formalize obliteration in the following way.

Proposition 5.13. *Obliteration $\mathbb{O}_S : \mathcal{V}(\mathcal{W}_{MS}) \rightarrow \mathcal{V}(\mathcal{W}_{MS})$ acts by*

$$(5.12) \quad \mathbb{O}_S(\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S, S_1, \dots, S_n)) := \gamma_{\mathcal{SO}, \mathcal{MO}}(T, \mathbf{1}, S_1, \dots, S_n) = \mathfrak{M}_{\mathbf{1}, S_1, \dots, S_n}^T(F),$$

where F is the morphological workspace $F = S \sqcup S_1 \sqcup \dots \sqcup S_n$.

Proof. We can view the feature bundle to be obliterated as a component S of a morphological workspace $F = S \sqcup \hat{F} = S \sqcup S_1 \sqcup \dots \sqcup S_n$, and we can assume that we start with a morphosyntactic tree of the form $\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S, S_1, \dots, S_n)$, where the components of the workspace F are inserted at the leaves of a syntactic tree T through the action of the morphosyntactic building operation

$$\sqcup \circ (\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots) \otimes \text{id} \circ \delta_{\underline{B}, \underline{\alpha}} \circ \Delta^\rho$$

restricted to the term where $\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots)$ acts on the primitive term $F \otimes \mathbf{1}$ of the coproduct.

In order to accommodate the obliteration of S , notice that the primitive part of the coproduct performs all the partitions of the workspace. In particular, there will be a term in the primitive part of the coproduct that is of the form $\hat{F} \otimes S = S_1 \sqcup \dots \sqcup S_n \otimes S$. We have $S_1 \sqcup \dots \sqcup S_n = \mathbf{1} \sqcup S_1 \sqcup \dots \sqcup S_n$, hence we can formally apply the insertion operation $\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots)$ to the term $\hat{F} \otimes S$ of the coproduct. This means performing the operation $\mathfrak{M}_{\mathbf{1}, S_1, \dots, S_n}^T$. $\square$

Note that the operation $\mathfrak{M}_{\mathbf{1}, S_1, \dots, S_n}^T$ is an analog here of the operation $\mathfrak{M}_{\mathbf{1}, S}$ used in syntax as a piece of the operation needed to model Internal Merge (see §1.4.3 of [17]).

Again, we write for simplicity the substitution $S \mapsto \mathbf{1}$ in the first position of $(S, S_1, \dots, S_n)$, but in fact it can be at any position S_ℓ corresponding to any leaf $\ell \in L(T)$.

Corollary 5.14. *We can reinterpret obliteration, defined as in (5.12), as the operation that maps $\mathbb{O}_S : \mathfrak{M}_{S, S_1, \dots, S_n}^T \mapsto \mathfrak{M}_{\mathbf{1}, S_1, \dots, S_n}^T$.*

Proof. This follows directly from Proposition 5.13 by writing in (5.12)

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S, S_1, \dots, S_n) = \mathfrak{M}_{S, S_1, \dots, S_n}^T(F)$$

for any workspace F that has a term of the form $S \sqcup S_1 \sqcup \cdots \sqcup S_n \otimes F'$, for some F' , in the coproduct $\Delta^\rho(F)$. $\square$

With this formulation of obliteration, one should further specify how to interpret the role of the leaf of the syntactic tree where the obliterated morphological S was inserted, when $S \mapsto \mathbf{1}$. One possibility is that the leaf ℓ with its label α_ℓ is also removed (with T replaced by the maximal full binary tree $T/^d\ell$ remaining after the cut of ℓ). The other possibility is that no morphology is inserted but the leaf α with its syntactic label α_ℓ is maintained. The mathematical formulation suggests that it should be the second case, since in the expression $\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \mathbf{1}, S_1, \dots, S_n)$ the term $\mathbf{1}$ means that no morphology is inserted leaving the leaf of T unchanged, and no operation $T/^d\ell$ is involved. This second case seems also preferable from the syntactic point of view, in terms of the No Complexity Loss principle for syntactic objects as discussed in §1.6.3 of [17].

5.3.1. *Obliteration of a subbundle of features.* More generally, obliteration can also apply to a part of a feature bundle as in the following example.

Example 5.15. Suppose that we start with a morphosyntactic tree

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S, S_1, \dots, S_n) = \mathfrak{M}_{S, S_1, \dots, S_n}^T(F)$$

for a morphological workspace $F = S \sqcup S_1 \sqcup \cdots \sqcup S_n \sqcup F'$, and with a morphological tree S of the form

$$(5.13) \quad \begin{array}{c} S = [\phi, \alpha, \beta, \gamma, \delta] \\ \swarrow \quad \searrow \\ [\phi, \alpha] \quad [\beta, \gamma, \delta] \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \phi \quad \alpha \quad \beta \quad [\gamma, \delta] \\ \quad \quad \quad \swarrow \quad \searrow \\ \quad \quad \quad \gamma \quad \delta \end{array}$$

that is inserted at a leaf v of a syntactic tree T , with $(B_v, \alpha_v) \in \Gamma_{SM}$ for $B_v = [\phi, \alpha, \beta, \gamma, \delta]$. Perform first a fission operation that transforms the tree above into

$$S_v = \begin{array}{c} v \\ \swarrow \quad \searrow \\ [\phi, \alpha] \quad [\beta, \gamma, \delta] \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \phi \quad \alpha \quad \beta \quad [\gamma, \delta] \\ \quad \quad \quad \swarrow \quad \searrow \\ \quad \quad \quad \gamma \quad \delta \end{array}$$

where now the insertion leaves are the vertices v_1 and v_2 below v with α_1, α_2 with $(B_i, \alpha_i) \in \Gamma_{SM}$ for $B_1 = [\phi, \alpha]$ and $B_2 = [\beta, \gamma, \delta]$ and one of the α_i equal to α_v , as discussed above. We can write this tree as

$$S_v = \gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_{B_1}, S_{B_2}),$$

with

$$(5.14) \quad \begin{array}{c} S_{B_1} = [\phi, \alpha] \quad \text{and} \quad S_{B_2} = [\beta, \gamma, \delta] \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \phi \quad \alpha \quad \beta \quad [\gamma, \delta] \\ \quad \quad \quad \swarrow \quad \searrow \\ \quad \quad \quad \gamma \quad \delta \end{array}$$

and the resulting morphosyntactic tree as

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T \circ_v \mathfrak{M}(\alpha_1, \alpha_2), S_{B_1}, S_{B_2}, S_1, \dots, S_n).$$

Suppose then that the bundle of features B_1 is the part that we want to obliterate and B_2 the part we want to keep. In the coproduct $\Delta^\rho(S_v)$, there is a term corresponding to the admissible cut that removes the edge connecting v to v_2 , which is of the form

$$\begin{array}{ccc} [\beta, \gamma, \delta] & \otimes & v \\ \beta \quad \gamma \quad \delta & & | \\ \beta \quad [\gamma, \delta] & & [\phi, \alpha] \\ \gamma \quad \delta & & \phi \quad \alpha \end{array}$$

There is then a term in the coproduct $\Delta^\rho(F)$ of the morphological workspace that is of the form

$$\begin{array}{ccc} [\beta, \gamma, \delta] & \sqcup S_1 \sqcup \dots \sqcup S_n & \otimes \quad v \quad \sqcup F' \\ \beta \quad \gamma \quad \delta & & | \\ \beta \quad [\gamma, \delta] & & [\phi, \alpha] \\ \gamma \quad \delta & & \phi \quad \alpha \end{array}$$

Applying

$$\sqcup \circ (\gamma_{\mathcal{SO}, \mathcal{MO}}(T', \dots) \otimes \text{id}) \circ \Delta^\rho$$

to this term of the coproduct then gives the morphosyntactic tree with the obliterated S_{B_1} .

Remark 5.16. For the same morphological tree of (5.13), where, as in Example 5.15, we want to remove the features ϕ and α while keeping β, γ , and δ , performing the fission operation is optional. Indeed, we might as well directly apply the coproduct to S and select the term of the form

$$\begin{array}{ccc} [\beta, \gamma, \delta] & \otimes & [\phi, \alpha, \beta, \gamma, \delta] \\ \beta \quad \gamma \quad \delta & & | \\ \beta \quad [\gamma, \delta] & & [\phi, \alpha] \\ \gamma \quad \delta & & \phi \quad \alpha \end{array}$$

When other components of the morphological workspace $S \sqcup_i S_i$ are also taken into consideration, one has a corresponding term in $\Delta^\rho(F)$ of the form

$$S' \sqcup_i S_i \otimes S/\rho S' \sqcup F'$$

for $F = S \sqcup_i S_i \sqcup F'$, and with

$$\begin{array}{ccc} S' := [\beta, \gamma, \delta] & \text{and} & S/\rho S' = [\phi, \alpha, \beta, \gamma, \delta] \\ \beta \quad \gamma \quad \delta & & | \\ \beta \quad [\gamma, \delta] & & [\phi, \alpha] \\ \gamma \quad \delta & & \phi \quad \alpha \end{array}$$

We can then form the morphosyntactic object

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T', S', S_1, \dots, S_n)$$

by applying

$$\sqcup \circ (\gamma_{\mathcal{SO}, \mathcal{MO}}(T', \dots) \otimes \text{id}) \circ \Delta^\rho$$

to this term of the coproduct.

The reason for introducing the fission operation in Example 5.15 is that it also allows us to obliterate *any other combinations of features*, as the following example shows.

Example 5.17. Suppose then that, in the same example of (5.13) we want instead to remove the features ϕ and γ and retain $[\alpha, \beta, \delta]$. This can now be done in a similar way, but it first requires using a fission operation that performs the separation of the sets of features $B_1 = [\phi, \gamma]$ and $B_2 = [\alpha, \beta, \delta]$, namely the operation $\Phi_{\emptyset, (B_1, B_2), \alpha_v}$ as in (5.7), where the procedure of Definition 5.6 for the construction of the $S_{B_1 \cup A}$ and $S_{B_2 \cup A}$ gives a resulting tree

$$S_v = \begin{array}{c} v \\ \swarrow \quad \searrow \\ S_{B_1} \quad S_{B_2} \end{array} = \begin{array}{c} v \\ \swarrow \quad \searrow \\ [\phi, \gamma] \quad [\alpha, \beta, \delta] \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \phi \quad \gamma \quad \alpha \quad [\beta, \delta] \\ \quad \quad \quad \swarrow \quad \searrow \\ \quad \quad \quad \beta \quad \delta \end{array}$$

The coproduct Δ^ρ will then produce a term $S_{v_2} \sqcup_i S_i \otimes S_v / {}^\rho S_{v_2}$ where

$$S_{v_2} = [\alpha, \beta, \delta] \quad , \quad \text{and} \quad S_v / {}^\rho S_{v_2} = \begin{array}{c} v \\ | \\ [\phi, \gamma] \\ \swarrow \quad \searrow \\ \phi \quad \gamma \end{array} .$$

We then apply the insertion $\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots)$ to this term of the coproduct, as in the previous example.

Cases like Example 5.15 and Example 5.17 are suitable for modeling situations where the presence of certain other features cause some of the features to be obliterated. This would mean that, in Externalization, a (language-dependent) filtering selects between the morphosyntactic structure before or after the obliteration operation, depending on the combination of features present in the feature bundle before obliteration. This type of filtering suggests a formulation in terms of coloring algorithms where certain adjacent combinations of colors are ruled out, as mentioned in Remark 5.11. We will not elaborate on this further in the present paper, as filtering in Externalization needs to be modeled separately.

Remark 5.18. The case discussed above, where obliteration targets a subbundle of features, as in Example 5.15 or Example 5.17, may be thought of either as Obliteration (though applied to only a part of the morphological tree) or as a case of Impoverishment. The difference between Impoverishment and Obliteration is sometimes described by the fact that in the first case there is still a realization of an existent feature bundle, which just does not include the normal features being realized, whereas Obliteration completely disallows any realization of a morphological node (and hence that node/feature bundle must have been completely deleted). We can also make the distinction in terms of whether a bundle or subbundle of features is completely obliterated, or whether a trace of it is maintained, either referring to the first as a form of Obliteration and the second as Impoverishment, or to both as two different forms of Impoverishment. We discuss these different cases in §5.4. We will give in Proposition 5.20 a comparative formulation, viewing these possibilities as different cases of Impoverishment.

5.4. Impoverishment. Impoverishment involves the obliteration of a piece of the feature bundle, as in the cases discussed in Examples 5.15 and 5.17, and in more general cases discussed below.

This occurs when there is reason to think multiple features (e.g., person and number) are normally realized, but in specific circumstances only some of those features are realized (e.g. only person, not number). An example of this can be seen in classical Arabic [14], as presented by [9].

Example 5.19. In the substantival declension of classical Arabic, certain substantives in the genitive indefinite do not express both superior and indefinite features, but rather express the default suffix. The following is a table of Arabic declensions.²

	NI	GI	AI	ND	GD	AD
raġul- ‘man’	-u-n	-i-n	-a-n	-u	-i	-a
riġāl- ‘men’	-u-n	-i-n	-a-n	-u	-i	-a
hāšim- ‘Hashim’	-u-n	-i-n	-a-n			
hārūn- ‘Aaron’	-u	-a	-a			
madāʔin- ‘cities’	-u	-a	-a	-u	-i	-a

The realization of *hārūn* ‘Aaron’ and *madāʔin* ‘cities’ in the genitive indefinite as compared to the other substantives as well as the genitive definite case is quite different—instead of being realized as *-i-n*, it is realized as *-a*. Positing /i/ as the vocabulary item for [+oblique], and /a/ as the elsewhere vocabulary item, as well as /n/ as the vocabulary item for [-definite] and $\emptyset$ as the elsewhere for definiteness, it appears that these two features ([+oblique] and [-definite]) have been impoverished and then realized as their elsewhere forms (/a/ and $\emptyset$).

Since we have conceptualized obliteration in Proposition 5.13 in terms of the coproduct Δ^ρ , as well as the cases discussed in Examples 5.15 and 5.17, the general form of impoverishment should also be a combination of fission and coproducts, to separate out the subtrees of features to be removed or kept, and to actually remove (as in obliteration) the unwanted part.

In our model, this would again correspond to fissioning the feature bundle into the piece to be obliterated and the piece to remain, and then obliterating that part of the feature bundle. If we view impoverishment as modeling cases where spell-out at a terminal node, by vocabulary insertion determined by specific features, is blocked by other less specific vocabulary items, then the difference with obliteration can be seen as maintaining the presence of certain features (and their possible interaction with other features, for instance in terms of determining filtering in Externalization), but no longer making the impoverished features available at the leaves (e.g., for vocabulary insertion). This indicates that, in a case like the tree of (5.13), if α and ϕ are the features to be retained and β, γ, δ are those targeted by Impoverishment, it is the term

$$\begin{array}{c}
 [\phi, \alpha, \beta, \gamma, \delta] \\
 | \\
 [\phi, \alpha] \\
 \wedge \\
 \phi \quad \alpha
 \end{array}$$

²N, G and A indicate nominative, genitive and accusative cases, respectively, while I and D indicate indefinite and definite.

in the right-channel of the coproduct

$$\begin{array}{c} [\beta, \gamma, \delta] \otimes [\phi, \alpha, \beta, \gamma, \delta] \\ \begin{array}{cc} \widehat{\beta} & \widehat{[\gamma, \delta]} \\ \gamma & \delta \end{array} \quad \begin{array}{c} | \\ [\phi, \alpha] \\ \widehat{\phi} \quad \alpha \end{array} \end{array}$$

that we want to use in replacement of the original morphological tree S of (5.13) rather than the term

$$\begin{array}{c} [\phi, \alpha] \\ \widehat{\phi} \quad \alpha \end{array}$$

in the left-channel of the term of the coproduct of the form

$$\begin{array}{c} [\phi, \alpha] \otimes [\phi, \alpha, \beta, \gamma, \delta] . \\ \begin{array}{cc} \widehat{\phi} & \alpha \end{array} \quad \begin{array}{c} | \\ [\beta, \gamma, \delta] \\ \widehat{\beta} \quad \widehat{[\gamma, \delta]} \\ \gamma \quad \delta \end{array} \end{array}$$

This requires moving a term from the right-channel of the coproduct to the workspace and then to the left-channel of a second application of the coproduct. This can be achieved, for morphological workspaces by first acting with the simplest Hopf algebra Markov chain $\sqcup \circ \Delta^\rho$. Applied to a workspace F , it generates a sum of terms, one of which is of the form $S_{B_2} \sqcup S_1 \sqcup \dots \sqcup S_n \sqcup S/\rho S_{B_2}$, with S_{B_1}, S_{B_2} as in (5.14), for $F = S \sqcup S_1 \sqcup \dots \sqcup S_n$.

(A similar situation arises the case of Internal Merge in syntax, where the extracted T_v is first deposited in the workspace and then merged with T/T_v .)

We can formalize the procedure described above in the following way.

Proposition 5.20. *The impoverishment operations $\mathbb{I}_{B \subset B_v} : \mathcal{V}(\mathcal{W}_{\mathcal{MS}}) \rightarrow \mathcal{V}(\mathcal{W}_{\mathcal{MS}})$ and $\mathbb{I}_{B_v/B} : \mathcal{V}(\mathcal{W}_{\mathcal{MS}}) \rightarrow \mathcal{V}(\mathcal{W}_{\mathcal{MS}})$ have two cases:*

- (1) *Obliteration of a subbundle of features (as discussed in §5.3.1).*
- (2) *Obliteration of a subbundle of features that maintains their trace (as outlined above).*

In the first case, for $B \subset B_v$ at the root v of an extended morphological tree S , the operation $\mathbb{I}_{B \subset B_v}$ acts as

$$(5.15) \quad \mathbb{I}_{B \subset B_v}(\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S, S_1, \dots, S_n)) = \gamma_{\mathcal{SO}, \mathcal{MO}}(T, S_{B'}, S_1, \dots, S_n),$$

where $B_v = B \sqcup B'$ and $S_{B'}$ is the subtree of the fission of S according to $B_v = B \sqcup B'$.

This impoverishment operation can be equivalently described as mapping

$$(5.16) \quad \mathbb{I}_{B \subset B_v} : \mathfrak{M}_{S, S_1, \dots, S_n}^T \mapsto \mathfrak{M}_{S_{B'}, S_1, \dots, S_n}^T .$$

The second case is similar, but of the form

$$(5.17) \quad \mathbb{I}_{B_v/B}(\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S, S_1, \dots, S_n)) = \gamma_{\mathcal{SO}, \mathcal{MO}}(T, \mathcal{F}_v \Phi_{A, (B, B')} / S_{B \cup A}, S_1, \dots, S_n),$$

where $\mathcal{F}_v \Phi_{A, (B, B')}$ is a fission of S with $S_{B \cup A}$ and $S_{B' \cup A}$ the two fissioned subtrees, followed by fusion $\mathcal{F}_v$ at its syntactic root vertex v . This is equivalent to the formulation

$$(5.18) \quad \mathbb{I}_{B_v/B} : \mathfrak{M}_{S, S_1, \dots, S_n}^T \mapsto \mathfrak{M}_{\mathcal{F}_v \Phi_{A, (B, B')} / S_{B \cup A}, S_1, \dots, S_n}^T .$$

Finally, we observe that the Obliteration and Impoverishment operations are not additional independent operations, but are obtainable from the fission and fusion operations and the coproduct Δ^ρ . Thus, the basic DM operations can be reduced to just fusion and fission.

Proposition 5.21. *The operations of Obliteration and Impoverishment are obtainable from combinations of fission, fusion, and the coproduct Δ^ρ .*

Proof. The case of Obliteration of a full feature bundle is already described in the proof of Proposition 5.13 in terms of the primitive part of the coproduct, so we focus on Impoverishment. For $B_v \setminus A = B \sqcup B'$ will use the shorthand notation for the fission operation:

$$S \mapsto \mathfrak{M}_\Phi^{\text{morph}}(S_{B \cup A}, S_{B' \cup A}) := \gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_v, \alpha), S_{B \cup A}, S_{B' \cup A}),$$

and for the composition of a fission and a fusion

$$S \mapsto \mathfrak{M}_{\mathcal{F}\Phi}^{\text{morph}}(S_{B \cup A}, S_{B' \cup A}) := \mathcal{F}_v(\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_v, \alpha), S_{B \cup A}, S_{B' \cup A})),$$

where $\mathcal{F}_v$ denotes the term of the fusion operation $\mathcal{F}$ that applies at the root vertex v of $\mathfrak{M}(\alpha_v, \alpha)$, which acquires the label (B_v, α_v) .

Note that this second operation, consisting of the composition of a fission and a fusion, does not in general give back S , because the two fissioned terms $S_{B \cup A}, S_{B' \cup A}$ do not in general satisfy $S = \mathfrak{M}^{\text{morph}}(S_{B \cup A}, S_{B' \cup A})$, see Example 5.7.

The first transformation $\mathbb{I}_{B \subset B_v}$ can be seen, in terms of building operations acting on morphological workspaces to assemble morphosyntactic objects, as the transformation that starts with a morphological workspace of the form $F = S \sqcup \sqcup_i S_i \sqcup F'$, and proceeds as

$$\begin{aligned} F &\xrightarrow{\Phi_S} \Phi_S(F) = \mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) \sqcup_i S_i \sqcup F' \\ &\xrightarrow{\Delta^\rho} S_{B'} \sqcup \mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) /^\rho S_{B'} \sqcup_i S_i \otimes F' + \text{other terms} \\ &\xrightarrow{\sqcup} S_{B'} \sqcup \mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) /^\rho S_{B'} \sqcup_i S_i \sqcup F' + \text{other terms}, \end{aligned}$$

where we write Φ_S for a fission operation that targets the component S of the workspace F . We then proceed with the new workspace

$$\tilde{F} := S_{B'} \sqcup \mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) /^\rho S_{B'} \sqcup_i S_i \sqcup F'$$

with

$$\begin{aligned} \tilde{F} &\xrightarrow{\Delta^\rho} (S_{B'} \sqcup_i S_i) \otimes (\mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) /^\rho S_{B'} \sqcup F') + \text{other terms} \\ &\xrightarrow{\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots) \otimes \text{id}} \gamma_{\mathcal{SO}, \mathcal{MO}}(T, S_{B'}, S_1, \dots, S_n) \otimes (\mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) /^\rho S_{B'} \sqcup F') \\ &\xrightarrow{\sqcup} \gamma_{\mathcal{SO}, \mathcal{MO}}(T, S_{B'}, S_1, \dots, S_n) \sqcup \mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) /^\rho S_{B'} \sqcup \sqcup F' \end{aligned}$$

where

$$\gamma_{\mathcal{SO}, \mathcal{MO}}(T, S_{B'}, S_1, \dots, S_n)$$

is the resulting morphosyntactic object formed, while

$$\mathfrak{M}_\Phi^{\text{morph}}(S_B, S_{B'}) /^\rho S_{B'} \sqcup F'$$

is the remaining discarded morphological material, that remains available for other morphosyntactic constructions.

The second case is similar. Again starting with a workspace of the form $F = S \sqcup \sqcup_i S_i \sqcup F'$ we proceed in the following way:

$$\begin{aligned} F &\xrightarrow{\mathcal{F}_v \circ \Phi_S} \Phi_S(F) = \mathfrak{M}_{\mathcal{F}\Phi}^{\text{morph}}(S_{B \cup A}, S_{B' \cup A}) \sqcup_i S_i \sqcup F' \\ &\xrightarrow{\sqcup \circ \Delta^\rho} S_{B \cup A} \sqcup_i S_i \sqcup \mathfrak{M}_{\mathcal{F}\Phi}^{\text{morph}}(S_{B \cup A}, S_{B' \cup A}) /^\rho S_{B \cup A} \sqcup F' + \text{other terms} \end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\Delta^\rho} (\mathfrak{M}_{\mathcal{F}\Phi}^{\text{morph}}(S_{B\cup A}, S_{B'\cup A}) / {}^\rho S_{B\cup A} \sqcup_i S_i) \otimes (S_{B'\cup A} \sqcup F') + \text{other terms} \\
& \xrightarrow{\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \dots) \otimes \text{id}} \gamma_{\mathcal{SO}, \mathcal{MO}}(T, \mathfrak{M}_{\mathcal{F}\Phi}^{\text{morph}}(S_{B\cup A}, S_{B'\cup A}) / {}^\rho S_{B\cup A}, S_1, \dots, S_n) \otimes (S_{B\cup A} \sqcup F') \\
& \xrightarrow{\sqcup} \gamma_{\mathcal{SO}, \mathcal{MO}}(T, \mathfrak{M}_{\mathcal{F}\Phi}^{\text{morph}}(S_{B\cup A}, S_{B'\cup A}) / {}^\rho S_{B\cup A}, S_1, \dots, S_n) \sqcup S_{B\cup A} \sqcup F',
\end{aligned}$$

where $\gamma_{\mathcal{SO}, \mathcal{MO}}(T, \mathfrak{M}_{\mathcal{F}\Phi}^{\text{morph}}(S_{B\cup A}, S_{B'\cup A}) / {}^\rho S_{B\cup A}, S_1, \dots, S_n)$ is the resulting morphosyntactic object and $S_{B\cup A} \sqcup F'$ is the discarded morphological material that remains available for further structure-building operations. $\square$

Remark 5.22. The form (5.17), (5.18) of the Impoverishment operation allows for implementing in our model the insertion of the unmarked feature that is relevant to various examples of Impoverishment. We will be discussing this more in detail elsewhere.

6. THE MOVABLE BOUNDARY OF MORPHOSYNTAX

The description of the fundamental operations of Distributed Morphology given above suggests that we should consider them as operations acting on the set $\mathcal{L}_{\mathcal{SO}, \mathcal{MO}}$ of (4.4) of morphosyntax building operations. Indeed, the characterization of fusion and fission as given in Proposition 5.3 and Proposition 5.9 allows us to identify fusion and fission with transformations

$$\begin{aligned}
& \mathcal{F}_{\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)} : \mathcal{L}_{\mathcal{SO}, \mathcal{MO}} \rightarrow \mathcal{L}_{\mathcal{SO}, \mathcal{MO}} \\
(6.1) \quad & \mathcal{F}_{\gamma_{\mathcal{SO}, \mathcal{MO}}(\mathfrak{M}(\alpha_1, \alpha_2), S_1, S_2)} : \mathfrak{M}_{S_1, S_2, \dots, S_n}^T \mapsto \mathfrak{M}_{S_{12}, \dots, S_n}^{T/c\mathfrak{M}(\alpha_1, \alpha_2)},
\end{aligned}$$

and

$$\begin{aligned}
& \Phi_{A, (B_1, B_2), \alpha} : \mathcal{L}_{\mathcal{SO}, \mathcal{MO}} \rightarrow \mathcal{L}_{\mathcal{SO}, \mathcal{MO}} \\
(6.2) \quad & \Phi_{A, (B_1, B_2), \alpha} : \mathfrak{M}_{S_1, \dots, S_\ell, \dots, S_n}^T \mapsto \mathfrak{M}_{S_1, \dots, S_\ell, B_1 \cup A, S_\ell, B_2 \cup A, \dots, S_n}^{T \circ \ell \mathfrak{M}(\alpha_\ell, \alpha)}
\end{aligned}$$

In a similar way, the Impoverishment or Obliteration operations of DM can be seen (up to composition with fission operations, as in Example 5.17 above) as transformations of $\mathcal{L}_{\mathcal{SO}, \mathcal{MO}} \rightarrow \mathcal{L}_{\mathcal{SO}, \mathcal{MO}}$ mapping

$$(6.3) \quad \mathfrak{M}_{S_1, \dots, S_\ell, \dots, S_n}^T \mapsto \mathfrak{M}_{S_1, \dots, S'_\ell, \dots, S_n}^T$$

where $S'_\ell \subset S_\ell$ is one of the two accessible terms immediately below the root of S_ℓ . Thus, we can view DM operations as a semigroup action on the set $\mathcal{L}_{\mathcal{SO}, \mathcal{MO}}$ obtained by arbitrary compositions of the generators (6.1), (6.2), (6.3).

Definition 6.1. *The Distributed Morphology semigroup $\mathcal{S}_{DM}$ is the semigroup generated by the operations (6.1), (6.2), (6.3), acting on the set $\mathcal{L}_{\mathcal{SO}, \mathcal{MO}}$. We refer to the subsemigroup generated by (6.1), (6.2) as the post-syntactic semigroup $\mathcal{S}_{PS} \subset \mathcal{S}_{DM}$.*

The main effect of the action of the semigroup $\mathcal{S}_{DM}$ on the set $\mathcal{L}_{\mathcal{SO}, \mathcal{MO}}$ is to render the boundary between syntax and morphology in the construction of morphosyntactic trees flexible, or movable (via the action of the generators (6.1), (6.2) and the post-syntactic semigroup $\mathcal{S}_{PS}$), along with the possibility of dropping some morphological features as effect of the generator (6.3). We focus here on the action of the post-syntactic semigroup $\mathcal{S}_{PS}$.

We refer to the dynamics implemented by these transformations as post-syntactic because it relies on formed syntactic objects and acts on the operations that interface syntax with morphology, hence they are not part of the computational structure of syntax, rather they are properties of the interface with morphology.

In a morphosyntactic tree, the two generators (6.1), (6.2) of $\mathcal{S}_{PS}$, representing the fusion and fission operations, respectively move upward or downward the vertices where the boundary between syntax and morphology occurs. In fusion, a syntactic vertex above the morphological insertions becomes the place where the morphological insertion takes place, while in fission a vertex of morphological insertion at a leaf of a syntactic tree becomes an interior syntactic vertex. In terms of the action on $\mathcal{L}_{SO,MO}$ this change is achieved by altering the assembly procedure of the morphosyntactic structure (replacing an element of $\mathcal{L}_{SO,MO}$, which is one such assembly procedure, with another one).

In Externalization, a filter can restrict the range of applicability of these transformations, limiting (in a language depended way) the amount of flexibility in the boundary between morphology and syntax, with polysynthetic languages (like Inuktitut) at one extreme, where the boundary can be significantly pushed upward into syntax, effectively absorbing syntax into word formation and morphology, and the most analytic languages (like Vietnamese) at the opposite extreme, where the boundary is pushed all the way downward, and with intermediate possibilities, for example agglutinating languages (like Swahili), fusional languages (like Semitic languages), oligosynthetic (like Nahuatl).

6.1. Additional remarks. We briefly discuss some additional remarks about the construction presented here. Note that the construction of morphosyntactic objects discussed in §4 would still work if one wants to consider more general forms of morphological trees that also involve higher valency vertices: the syntactic part of the tree would remain binary, while the morphological parts would include both binary and higher valence vertices. One may worry then that the presence of non-binary trees in the morphological part would affect the possibility of moving the morpho-syntactic boundary via the fusion and fission operations, but this is not the case. The fusion operation transforms a binary syntactic vertex into a binary morphological vertex, which would still be available. The fission operations are design to split a bundle of morphological features into *two* parts and construct the two corresponding morphological trees using the set theoretic splitting of the feature bundle and the original morphological tree structure, as described in Definition 5.6. This will in any case generate a new syntactic vertex that is necessarily binary, as required for syntax, even if the resulting two morphological trees produced following the algorithm of Definition 5.6 may have higher valence vertices. Thus, the boundary between syntax and morphology would remain movable even if morphological structures are realized by trees with higher valence vertices (as used in the case of feature geometry). Our choice to represent all morphological tree as binary is motivated by optimality (significant simplification of the algebraic structure generating them).

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REFERENCES

- [1] Arregi, K. and Nevins, A. (2012). *Morphotactics: Basque auxiliaries and the structure of spellout*. Springer Dordrecht.
- [2] Baunaz, L., Haegeman, L., De Clercq, K., and Lander, E. (2018a). *Exploring nanosyntax*. Oxford University Press.
- [3] Baunaz, L., Lander, E., De Clercq, K., and Haegeman, L. (2018b). Nanosyntax: the basics. *Oxford Studies in Comparative Syntax*, pages 3–56. Publisher: Oxford University Press.
- [4] Bobaljik, J. D. (1997). If the head fits...: on the morphological determination of Germanic syntax. Publisher: Walter de Gruyter, Berlin/New York Berlin, New York.
- [5] Caha, P. (2009). The nanosyntax of case. Publisher: Universitetet i Tromsø.
- [6] Coon, J. and Keine, S. (2021). Feature gluttony. *Linguistic inquiry*, 52(4):655–710. Publisher: MIT press journals-info@ mit. edu.
- [7] Cowper, E. and Hall, D. C. (2002). The syntactic manifestation of nominal feature geometry. In *Proceedings of the 2002 annual conference of the Canadian Linguistic Association*, pages 55–66. Canadian Linguistic Association Toronto.
- [8] Cowper, E. A. (2005). The geometry of interpretable features: Infl in English and Spanish. *Language*, 81(1):10–46. Publisher: Linguistic Society of America.
- [9] Embick, D. and Noyer, R. (2007). Distributed morphology and the syntax—morphology interface.
- [10] Halle, M. and Marantz, A. (1993). Distributed morphology and the pieces of inflection. In Hale, K. and Keyser, S., editors, *The view from building 20*, pages 111–176. The MIT Press, Cambridge, MA.
- [11] Halle, M. and Marantz, A. (1994). Some key features of distributed morphology. In Carnie, A., Harley, H., and Bures, T., editors, *MITWPL 21*, pages 275–288.
- [12] Harbour, D. (2008). Discontinuous agreement and the syntax-morphology interface. *Phi theory: Phi-features across modules and interfaces*, pages 185–220. Publisher: Oxford University Press Oxford.
- [13] Harley, H. E. and Ritter, E. (2002). Person and number in pronouns: A feature-geometric analysis. *Language*, 78:482 – 526.
- [14] Haywood, J. and Nahmad, H. (1965). *Arabic grammar of the written language*. Lund Humphries.
- [15] Hewett, M. (2023). Allomorphy in Semitic discontinuous agreement: Evidence for a modular approach to postsyntax. *Natural Language & Linguistic Theory*, 41(3):1091–1145. Publisher: Springer.
- [16] Kleine, S. and Müller, G. (preprint). Impoverishment.
- [17] Marcolli, M., Chomsky, N., and Berwick, R. C. (2025a). *Mathematical Structure of Syntactic Merge: An Algebraic Model for Generative Linguistics*. MIT Press.
- [18] Marcolli, M., Huijbregts, R., and Larson, R. K. (2025b). Hypermagmas and colored operads: heads, phases, and theta roles.
- [19] Marcolli, M., Larson, R., and Huijbregts, R. (2024). Extension Condition “violations” and Merge optimality constraints.
- [20] Marcolli, M. and Larson, R. K. (2025). Theta theory: operads and coloring. *arXiv:2503.06091*.
- [21] Martinović, M. (2022). Feature geometry and head splitting in the Wolof clausal periphery. *Linguistic Inquiry*, 54(1):79–116. Publisher: MIT press journals-info@ mit. edu.

- [22] McGinnis, M. (2007). Phi-feature competition in morphology and syntax. Publisher: Oxford University Press.
- [23] Nevins, A. (2016). Lectures on postsyntactic morphology. *Ms. University College London*.
- [24] Noyer, R. (1997). *Features, Positions and Affixes in Autonomous Morphological Structure*. Garland Publishing.
- [25] Starke, M. (2009). Nanosyntax: A short primer to a new approach to language. *Nordlyd*, 36(1):1–6.
- [26] Svenonius, P. (2006). Case alternations and the Icelandic passive and middle. *Passives and impersonals in European languages*. Publisher: Citeseer.
- [27] Tosco, M. (2007). Feature-geometry and diachrony: the development of the subject clitics in Cushitic and Romance. *Diachronica*, 24(1):119–153. Publisher: John Benjamins.
- [28] Watson, J. C. (1993). *A syntax of anānī Arabic*, volume 13. Otto Harrassowitz Verlag.
- [29] Xu, Y. (2018). A double-person feature analysis of Algonquian first person plurals. In *Proceedings of WCCFL*, volume 35.

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